Signorini's Problem

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February 26, 2026

Contents

Abstract

This is the abstract.[1]

1 Introduction

This is the Introduction.

2 Mathematical Framework

This section introduces the basic notation used throughout the mathematical framework and the later analysis of Signorini's problem. Our goal is to fix a consistent language for bulk quantities, boundary quantities, and tensor-valued fields before introducing weak derivatives and Sobolev spaces. Unless stated otherwise, let $\Omega \subset \mathbb{R}^d$, $d \geq 1$, be a nonempty open set. Whenever trace operators or boundary spaces are used, we additionally assume that Ω is a bounded Lipschitz domain (see e.g. [1, Section 2.3]) and write $\Gamma := \partial\Omega$.

2.1 Scalars, Vectors and Matrices

The purpose of this subsection is mainly to introduce the notation used throughout this report. For $m, n \geq 1$, we write $\mathbb{R}^{m\times n}$ for the space of real $m \times n$ matrices and for vectors $u, v \in \mathbb{R}^m$, we denote by

$$u \cdot v := \sum_{j=1}^{m} u_j\, v_j, \qquad |u| := \sqrt{u \cdot u}$$

the Euclidean inner product and norm. For matrices $A, B \in \mathbb{R}^{m\times n}$, we use the Frobenius product

$$A : B := \sum_{j=1}^{m}\sum_{k=1}^{n} A_{jk}B_{jk}, \qquad |A| := \sqrt{A : A}.$$

The space $\mathrm{Sym}^2(\mathbb{R}^d)$ of symmetric second order tensors is

$$\mathrm{Sym}^2(\mathbb{R}^d) := \left\{ s \in \mathbb{R}^d \otimes \mathbb{R}^d \;\middle|\; s^\top = s \right\}.$$

We will interpret these tensors as symmetric matrices in $\mathbb{R}^{d\times d}$ in the canonical basis of $\mathbb{R}^d$. Indeed,

$$\mathrm{Sym}^2(\mathbb{R}^d) \cong \left\{ S \in \mathbb{R}^{d\times d} \;\middle|\; S^\top = S \right\}.$$

Furthermore, if $\mathcal{A} : \Omega \to \mathcal{L}(\mathrm{Sym}^2(\mathbb{R}^d); \mathrm{Sym}^2(\mathbb{R}^d))$ is a fourth order tensor and $s \in \mathrm{Sym}^2(\mathbb{R}^d)$, we use the *double contraction*

$$(\mathcal{A}(x) : s)_{ij} := \sum_{k=1}^{d}\sum_{l=1}^{d} \mathcal{A}_{ijkl}(x)s_{kl}, \qquad s : \mathcal{A}(x) : s := \sum_{i=1}^{d}\sum_{j=1}^{d}\sum_{k=1}^{d}\sum_{l=1}^{d} \mathcal{A}_{ijkl}(x)s_{ij}s_{kl}.$$

2.2 L^p-Spaces

Even assuming the reader is already familiar with the concept of L^p-spaces, the definitions and notations are listed below. We begin with the L^p-spaces on Ω as spaces of equivalence classes of p-integrable functions whose difference vanishes almost everywhere on Ω. Although this is the most correct way to define L^p-spaces, we will subsequently always assume an actual function by choosing an arbitrary representative of a class. The assumed measure on Ω here is the Lebesgue measure in $\mathbb{R}^d$. We omit the case $p = \infty$.

Definition 2.1 (L^p-Spaces)**.** *Let $m \geq 1$ and $1 \leq p < \infty$. We define*

$$L^p(\Omega; \mathbb{R}^m) := \left\{ u : \Omega \to \mathbb{R}^m \text{ measurable} \;\middle|\; \int_\Omega |u|^p \,\mathrm{d}x < \infty \right\} \Big/ \sim,$$

where $u \sim v$ if and only if $u = v$ almost everywhere in Ω. For some equivalence class $[u] \in L^p(\Omega; \mathbb{R}^m)$ we take an arbitrary representative and write $u = [u]$. If $m = 1$, we abbreviate

$L^p(\Omega) := L^p(\Omega; \mathbb{R}^1)$. *Moreover, we equip* $L^p(\Omega; \mathbb{R}^m)$ *with the norm*

$$\|u\|_{0,p,\Omega} := \left(\int_\Omega |u|^p \, \mathrm{d}x \right)^{1/p}, \qquad u \in L^p(\Omega; \mathbb{R}^m).$$

The notation $\|\cdot\|_{0,p,\Omega}$ will be more clear after Sobolev spaces are introduced. Next, it is necessary to define these spaces for boundary parts of Ω as well. This is done analogously, with the difference that we now consider a surface or boundary integral and accordingly also the Lebesgue measure on $\mathbb{R}^{d-1}$.

Definition 2.2 (L^p-Spaces on Boundary Parts)**.** *Assume that* Ω *is a bounded Lipschitz domain and let* $\Gamma_0 \subset \Gamma$ *be measurable with respect to the surface measure on* Γ*. For* $m \geq 1$ *and* $1 \leq p < \infty$*, we define*

$$L^p(\Gamma_0; \mathbb{R}^m) := \left\{ g : \Gamma_0 \to \mathbb{R}^m \text{ measurable} \;\middle|\; \int_{\Gamma_0} |g|^p \, \mathrm{d}s < \infty \right\} \Big/ \sim,$$

where $g \sim h$ *if and only if* $g = h$ *almost everywhere on* Γ_0*. For some equivalence class* $[g] \in L^p(\Gamma_0; \mathbb{R}^m)$ *we take an arbitrary representative and write* $g = [g]$*. If* $m = 1$*, we abbreviate* $L^p(\Gamma_0) := L^p(\Gamma_0; \mathbb{R}^1)$*. Moreover, we equip* $L^p(\Gamma_0; \mathbb{R}^m)$ *with the norm*

$$\|g\|_{0,p,\Gamma_0} := \int_{\Gamma_0} |g|^p \, \mathrm{d}x, \qquad g \in L^p(\Gamma_0; \mathbb{R}^m).$$

Moreover, we also need an L^p-notation for matrix-valued fields $A : \Omega \to \mathrm{Sym}^2(\mathbb{R}^d)$.

Definition 2.3 (L^p-Spaces for Matrix-Valued Fields)**.** *Let* $1 \leq p < \infty$*. We define*

$$L^p(\Omega; \mathrm{Sym}^2(\mathbb{R}^d)) := \left\{ A : \Omega \to \mathrm{Sym}^2(\mathbb{R}^d) \text{ measurable} \;\middle|\; \int_\Omega |A|^p \, \mathrm{d}x < \infty \right\} \Big/ \sim,$$

where $A \sim B$ *if and only if* $A = B$ *almost everywhere on* Ω*. For some equivalence class* $[A] \in L^p(\Omega; \mathrm{Sym}^2(\mathbb{R}^d))$*, we take an arbitrary representative and write* $A = [A]$*. Moreover, we equip* $L^p(\Omega; \mathrm{Sym}^2(\mathbb{R}^d))$ *with the norm*

$$\|A\|_{0,p,\Omega} := \int_\Omega |A|^p \, \mathrm{d}x, \qquad A \in L^p(\Omega; \mathrm{Sym}^2(\mathbb{R}^d)).$$

Finally, one particular L^p-space is of special interest because it is the only one that, together with its classical inner product, defines a Hilbert space. This is the case $p = 2$. We record this fact in the following proposition and simultaneously introduce a meaningful notation for its inner product and its norm.

Proposition 2.1 (L^2 is a Hilbert Space)**.** *For every* $m \geq 1$*, the space* $L^2(\Omega; \mathbb{R}^m)$ *is a Hilbert space with inner product*

$$(u, v)_{0,\Omega} := \int_\Omega u \cdot v \, \mathrm{d}x \qquad u, v \in L^2(\Omega; \mathbb{R}^m),$$

and the induced norm is $\|u\|_{0,\Omega} := \|u\|_{0,2,\Omega} = (u, u)_{0,\Omega}^{1/2}$*. If* Ω *is a bounded Lipschitz domain and* $\Gamma_0 \subset \Gamma$ *is measurable,* $L^2(\Gamma_0, \mathbb{R}^m)$ *is a Hilbert space and for* $g, h \in L^2(\Gamma_0, \mathbb{R}^m)$ *we analogously write*

$$(g, h)_{0,\Gamma_0} := \int_{\Gamma_0} g \cdot h \, \mathrm{d}s, \qquad \|g\|_{0,\Gamma_0} := \|g\|_{0,2,\Gamma_0} = (g, g)_{0,\Gamma_0}^{1/2}.$$

Moreover, $L^2(\Omega; \mathrm{Sym}^2(\mathbb{R}^d))$ *together with the inner product*

$$(A,\, B)_{0,\Omega} := \int_\Omega A : B \,\mathrm{d}x, \qquad A, B \in L^2(\Omega; \mathrm{Sym}^2(\mathbb{R}^d))$$

is a Hilbert space and the induced norm is

$$\|A\|_{0,\Omega} := \|A\|_{0,2,\Omega} = (A,\, A)_{0,\Omega}^{1/2}.$$

For later purpose, we also need to consider the subspace of locally integrable functions on Ω.

Definition 2.4 (Locally Integrable Functions)**.** *The space of* `locally integrable` *functions on* Ω *is defined as*

$$L^1_{\mathrm{loc}}(\Omega; \mathbb{R}^m) := \left\{ u : \Omega \to \mathbb{R}^m \text{ measurable} \,\middle|\, u|_K \in L^1(K; \mathbb{R}^m) \; \forall K \subset \Omega,\, K \text{ compact} \right\}.$$

For $m = 1$*, we denote* $L^1_{\mathrm{loc}}(\Omega) := L^1_{\mathrm{loc}}(\Omega; \mathbb{R}^1)$ *and the space* $L^1_{\mathrm{loc}}(\Omega; \mathrm{Sym}^2(\mathbb{R}^d))$ *is defined analogously.*

2.3 Test Functions & Distributional Derivatives

Before introducing Sobolev spaces, we first formalize a weak notion of differentiation. The reason is that the functions arising in variational formulations of boundary value problems are typically not smooth enough to admit classical derivatives in the pointwise sense. Nevertheless, one still wants to differentiate such functions in a meaningful way, at least under an integral sign. The key observation is that in integration by parts, derivatives can be shifted from an unknown function to a smooth test function with compact support. This suggests defining derivatives indirectly through their action on test functions. The resulting objects are distributions, and distributional derivatives extend classical derivatives whenever the latter exist. For the sake of completeness, we begin with the definitions of the classic notions of differentiation, which should already be familiar to the reader.

Definition 2.5 (Partial Derivative, Gradient, Jacobian and Divergence)**.** *Let* $u : \Omega \to \mathbb{R}$ *and* $F : \Omega \to \mathbb{R}^m$ *and* $A : \Omega \to \mathbb{R}^{m\times d}$ *be sufficiently regular maps.*

i) The `partial derivative` *of* u *with respect to* $\alpha \in \mathbb{N}_0^d$ *is*

$$\mathrm{D}^\alpha u := \frac{\partial^{|\alpha|} u}{\partial x_1^{\alpha_1} \cdots \partial x_d^{\alpha_d}} : \Omega \to \mathbb{R}.$$

ii) The `gradient` *of* u *is*

$$\nabla u := \left(\frac{\partial u}{\partial x_1}, \ldots, \frac{\partial u}{\partial x_d} \right)^\top : \Omega \to \mathbb{R}^d.$$

iii) The `Jacobian` *of* $F = (F_1, \ldots, F_m)^\top$ *is*

$$\mathrm{D}F := \left(\frac{\partial F_j}{\partial x_k} \right)_{1 \le j \le m}^{1 \le k \le d} : \Omega \to \mathbb{R}^{m\times d}.$$

iv) For $m = d$*, the* `divergence` *of* $F = (F_1, \ldots, F_d)^\top$ *is*

$$\operatorname{div} F := \sum_{j=1}^d \frac{\partial F_j}{\partial x_j} : \Omega \to \mathbb{R}.$$

v) The `divergence` *of* $A = (A_{jk})_{1\le j\le m}^{1\le k\le d}$ *is*

$$\operatorname{div} A := \left(\sum_{k=1}^{d} \frac{\partial A_{jk}}{\partial x_k}\right)_{1\le j\le m} : \Omega \to \mathbb{R}^m.$$

We collect continuous functions with continuous partial derivatives of the same order in the following well-known function spaces.

Definition 2.6 (Spaces of Continuously Differentiable Functions)**.** *We denote by* $\mathcal{C}^0(\Omega;\mathbb{R}^m)$ *the space of continuous functions* $u : \Omega \to \mathbb{R}^m$ *and by*

$$\mathcal{C}^k(\Omega;\mathbb{R}^m) := \left\{ u \in \mathcal{C}^0(\Omega;\mathbb{R}^m) \;\middle|\; |\alpha| \le k \;\Rightarrow\; \mathrm{D}^\alpha u \in \mathcal{C}^0(\Omega;\mathbb{R}^m) \right\}, \qquad k \ge 0,$$

the space of such functions that are continuously differentiable up to order $k \ge 0$*. Moreover,*

$$\mathcal{C}^\infty(\Omega;\mathbb{R}^m) := \bigcap_{k\ge 0} \mathcal{C}^k(\Omega;\mathbb{R}^m)$$

denotes the space of infinitely continuously differentiable functions. For $m = 1$*, we abbreviate*

$$\mathcal{C}^k(\Omega) := \mathcal{C}^k(\Omega;\mathbb{R}^1), \qquad \forall\, 0 \le k \le \infty$$

and note that

$$\mathcal{C}^k(\Omega;\mathbb{R}^m) = (\mathcal{C}^k(\Omega))^m, \quad \forall\, 0 \le k \le \infty,\, m \ge 1.$$

Finally, for $0 \le k \le \infty$*, we define*

$$\mathcal{C}^k(\Omega;\mathrm{Sym}^2(\mathbb{R}^d)) := \left\{ A = (A_{ij})_{i,j=1}^d : \Omega \to \mathrm{Sym}^2(\mathbb{R}^d) \;\middle|\; A_{ij} \in \mathcal{C}^k(\Omega)\ \forall 1 \le i,j \le d \right\}.$$

A particular subspace of the $k = \infty$ case is of special interest to us and the concept of distributional derivatives. This is the space of infinitely often continuously differentiable functions with compact support, also called *test functions*.

Definition 2.7 (Test Function)**.** *The space* $\mathcal{D}(\Omega;\mathbb{R}^m)$ *contains all infinitely often continuously differentiable functions with compact support in* Ω*, i.e.*

$$\mathcal{D}(\Omega;\mathbb{R}^m) = \{\varphi \in \mathcal{C}^\infty(\Omega;\mathbb{R}^m) \mid \operatorname{supp}(\varphi) \subset \Omega \text{ compact}\}$$

For $m = 1$*, we abbreviate* $\mathcal{D}(\Omega) := \mathcal{D}(\Omega;\mathbb{R}^1)$ *and note that*

$$\mathcal{D}(\Omega;\mathbb{R}^m) = (\mathcal{D}(\Omega))^m, \qquad \forall\, m \ge 1.$$

The space $\mathcal{D}(\Omega;\mathrm{Sym}^2(\mathbb{R}^d))$ *is defined analogously and functions of these spaces are called* `test functions`*.*

The space of test functions allows us to encode differentiation through integration by parts. This leads to distributions, which extend classical derivatives to nonsmooth functions and, more generally, to generalized functions.

Definition 2.8 (Distributions)**.** *The space of scalar-valued* `distributions` *on* Ω *is the topological dual space*

$$\mathcal{D}'(\Omega) := (\mathcal{D}(\Omega))'.$$

Thus, a scalar-valued distribution is a linear continuous functional $T : \mathcal{D}(\Omega) \to \mathbb{R}$. For $T \in \mathcal{D}'(\Omega)$ and $\varphi \in \mathcal{D}(\Omega)$, we denote the duality pairing by $\langle T, \varphi\rangle$. For $m \geq 1$, we define the space of vector-valued distributions componentwise by

$$\mathcal{D}'(\Omega;\mathbb{R}^m) := \left(\mathcal{D}'(\Omega)\right)^m.$$

For $T = (T_1, \dots, T_m) \in \mathcal{D}'(\Omega;\mathbb{R}^m)$ and $\varphi = (\varphi_1, \dots, \varphi_m) \in \mathcal{D}(\Omega;\mathbb{R}^m)$, we use the duality pairing

$$\langle T, \varphi\rangle := \sum_{j=1}^{m} \langle T_j, \varphi_j\rangle.$$

Analogously, the space $\mathcal{D}'(\Omega; \mathrm{Sym}^2(\mathbb{R}^d))$ is defined componentwise, and for

$$T = (T_{jk})_{j,k=1}^d \in \mathcal{D}'(\Omega; \mathrm{Sym}^2(\mathbb{R}^d)), \qquad \varphi = (\varphi_{jk})_{j,k=1}^d \in \mathcal{D}(\Omega; \mathrm{Sym}^2(\mathbb{R}^d)),$$

we use the duality pairing

$$\langle T, \varphi\rangle := \sum_{j=1}^{d}\sum_{k=1}^{d} \langle T_{jk}, \varphi_{jk}\rangle.$$

Definition 2.9 (Distributional Derivative)**.** *Let $T \in \mathcal{D}'(\Omega)$ and $\alpha \in \mathbb{N}_0^d$. The* `distributional derivative` $\mathrm{D}^\alpha T \in \mathcal{D}'(\Omega)$ *is defined by*

$$\langle \mathrm{D}^\alpha T, \varphi\rangle := (-1)^{|\alpha|} \langle T, \mathrm{D}^\alpha \varphi\rangle, \qquad \forall\, \varphi \in \mathcal{D}(\Omega).$$

Distributional derivatives of vector-valued and matrix-valued distributions are defined componentwise. In particular, for $T = (T_1, \dots, T_m) \in \mathcal{D}'(\Omega;\mathbb{R}^m)$, we define

$$\mathrm{D}^\alpha T := (\mathrm{D}^\alpha T_1, \dots, \mathrm{D}^\alpha T_m) \in \mathcal{D}'(\Omega;\mathbb{R}^m).$$

Analogously, for $T = (T_{jk})_{j,k=1}^d \in \mathcal{D}'(\Omega; \mathrm{Sym}^2(\mathbb{R}^d))$, we define

$$\mathrm{D}^\alpha T := (\mathrm{D}^\alpha T_{jk})_{j,k=1}^d \in \mathcal{D}'(\Omega; \mathrm{Sym}^2(\mathbb{R}^d)).$$

Using the componentwise definition of distributional derivatives, we can now extend the classical differential operators introduced above to distributions. For $1 \leq k \leq d$, let $e_k \in \mathbb{N}_0^d$ denote the k-th unit multi-index, i.e.

$$e_k := (0, \dots, 0, 1, 0, \dots, 0)^\top,$$

where the entry 1 is in the k-th position. Then D^{e_k} in the distributional sense corresponds to the first-order derivative in the x_k-direction.

Definition 2.10 (Distributional Gradient, Jacobian and Divergence)**.** *Let $T \in \mathcal{D}'(\Omega)$ and $U \in \mathcal{D}'(\Omega;\mathbb{R}^m)$ and $\Sigma \in \mathcal{D}'(\Omega; \mathrm{Sym}^2(\mathbb{R}^d))$.*

i) The `distributional gradient` *of T is*

$$\nabla T := (\mathrm{D}^{e_1} T, \dots, \mathrm{D}^{e_d} T)^\top \in \mathcal{D}'(\Omega;\mathbb{R}^d).$$

ii) The `distributional Jacobian` *of $U = (U_1, \dots, U_m)^\top$ is*

$$\mathrm{D}U := (\mathrm{D}^{e_k} U_j)_{1\leq j\leq m}^{1\leq k\leq d} \in \mathcal{D}'(\Omega;\mathbb{R}^{m\times d}).$$

iii) For $d = m$, the `distributional divergence` *of $U = (U_1, \ldots, U_d)^\top$ is*

$$\operatorname{div} U := \sum_{k=1}^{d} \mathrm{D}^{e_k} U_k \in \mathcal{D}'(\Omega).$$

iv) The `distributional divergence` *of $\Sigma = (\Sigma_{jk})_{j,k=1}^{d}$ is*

$$\operatorname{div} \Sigma := \left(\sum_{k=1}^{d} \mathrm{D}^{e_k} \Sigma_{jk} \right)_{1 \le j \le d} \in \mathcal{D}'(\Omega; \mathbb{R}^d).$$

At this stage, the above definitions are purely distributional and do not require any pointwise differentiability. In a later subsection, we will identify locally integrable functions with regular distributions and show that weak derivatives are precisely those distributional derivatives that are again induced by functions.

2.4 Regular Distributions & Weak Derivatives

The distributional framework introduced above becomes particularly useful once distributions are induced by locally integrable functions. These so-called regular distributions provide the link between generalized differentiation and the weak derivatives used later in Sobolev spaces and variational formulations. From now on let X denote one of the spaces $\mathbb{R}^m$, $m \ge 1$ or $\operatorname{Sym}^2(\mathbb{R}^d)$ $d \ge 2$ to save some writing work and recall that $(\cdot, \cdot)_{0,\Omega}$ denotes the inner product on each of the spaces $L^2(\Omega; X)$ as introduced in Proposition 2.1. On X we use the notation of $(\cdot, \cdot)_X$ as the inner product, i.e. Euclidean or Frobenius, respectively, as stated in Section 2.1.

Definition 2.11 (Regular and Singular Distributions). *A* `regular distribution` *is a distribution $T_f \in \mathcal{D}'(\Omega; X)$ induced by a locally integrable function $f \in L^1_{\mathrm{loc}}(\Omega; X)$ by*

$$\langle T_f, \varphi \rangle := \int_\Omega (f, \varphi)_X \, \mathrm{d}x, \qquad \forall\, \varphi \in \mathcal{D}(\Omega; X).$$

In particular, $L^1_{\mathrm{loc}}(\Omega; X)$ is embedded into $\mathcal{D}'(\Omega; X)$ via $f \mapsto T_f$ and its image is the space of regular distributions. Distributions that are not regular are called `singular`*.*

The concept of distributional derivatives applied to regular distributions is exactly the concept of weak derivatives.

Definition 2.12 (Weak Derivative). *Let $f \in L^1_{\mathrm{loc}}(\Omega; X)$ and $\alpha \in \mathbb{N}_0^d$. A locally integrable function $g \in L^1_{\mathrm{loc}}(\Omega; X)$ is called the* `weak derivative` *of f of order α if*

$$\int_\Omega (g, \varphi)_X \, \mathrm{d}x = (-1)^{|\alpha|} \int_\Omega (f, \mathrm{D}^\alpha \varphi)_X \, \mathrm{d}x \qquad \forall\, \varphi \in \mathcal{D}(\Omega; X).$$

In this case, we write $g = \mathrm{D}^\alpha f$.

Proposition 2.2. *Let $f, g \in L^1_{\mathrm{loc}}(\Omega; X)$. Then g is the weak derivative $\mathrm{D}^\alpha f$ of f if and only if*

$$\langle \mathrm{D}^\alpha T_f, \varphi \rangle = \langle T_g, \varphi \rangle \qquad \forall\, \varphi \in \mathcal{D}(\Omega; X),$$

or in other words $\mathrm{D}^\alpha T_f = T_g$ in $\mathcal{D}'(\Omega; X)$.

Proof. Assume first that $g = \mathrm{D}^\alpha f$ in the weak sense. Then for every $\varphi \in \mathcal{D}(\Omega; X)$, by definition of the distributional derivative and the weak derivative,

$$\langle \mathrm{D}^\alpha T_f,\, \varphi\rangle = (-1)^{|\alpha|}\,\langle T_f,\, \mathrm{D}^\alpha\varphi\rangle = (-1)^{|\alpha|}\int_\Omega (f,\, \mathrm{D}^\alpha\varphi)_X \,\mathrm{d}x = \int_\Omega (g,\, \varphi)_X \,\mathrm{d}x = \langle T_g,\, \varphi\rangle.$$

Hence $\mathrm{D}^\alpha T_f = T_g$ in $\mathcal{D}'(\Omega; X)$. Conversely, assume that $\mathrm{D}^\alpha T_f = T_g$ in $\mathcal{D}'(\Omega; X)$. Then for every test function $\varphi \in \mathcal{D}(\Omega; X)$,

$$\int_\Omega (g,\, \varphi)_X \,\mathrm{d}x = \langle T_g,\, \varphi\rangle = \langle \mathrm{D}^\alpha T_f,\, \varphi\rangle = (-1)^{|\alpha|}\,\langle T_f,\, \mathrm{D}^\alpha\varphi\rangle = (-1)^{|\alpha|}\int_\Omega (f,\, \mathrm{D}^\alpha\varphi)_X \,\mathrm{d}x.$$

Thus, g is the weak derivative $\mathrm{D}^\alpha f$ of f. ■

Proposition 2.3 (Differential Operators on Regular Distributions). *Let $u \in L^1_{\mathrm{loc}}(\Omega)$ and $F \in L^1_{\mathrm{loc}}(\Omega; \mathbb{R}^d)$ and $A \in L^1_{\mathrm{loc}}(\Omega; \mathrm{Sym}^2(\mathbb{R}^d))$.*

i) For every $\varphi \in \mathcal{D}(\Omega; \mathbb{R}^d)$,

$$\langle \nabla T_u,\, \varphi\rangle = -\langle T_u,\, \operatorname{div}\varphi\rangle = -\int_\Omega u \operatorname{div}\varphi \,\mathrm{d}x,$$

where $\operatorname{div}\varphi$ is understood in the classical sense. If the weak gradient ∇u exists in $L^1_{\mathrm{loc}}(\Omega; \mathbb{R}^d)$, then

$$\nabla T_u = T_{\nabla u} \qquad \text{in } \mathcal{D}'(\Omega; \mathbb{R}^d).$$

ii) For every $\varphi \in \mathcal{D}(\Omega)$,

$$\langle \operatorname{div} T_F,\, \varphi\rangle = -\langle T_F,\, \nabla\varphi\rangle = -\int_\Omega F \cdot \nabla\varphi \,\mathrm{d}x,$$

where $\nabla\varphi$ is understood in the classical sense. If, in addition, the weak divergence $\operatorname{div} F$ exists in $L^1_{\mathrm{loc}}(\Omega)$, then

$$\operatorname{div} T_F = T_{\operatorname{div} F} \qquad \text{in } \mathcal{D}'(\Omega).$$

iii) For every $\varphi \in \mathcal{D}(\Omega; \mathbb{R}^d)$,

$$\langle \operatorname{div} T_A,\, \varphi\rangle = -\int_\Omega A : \mathrm{D}\varphi \,\mathrm{d}x,$$

where $\mathrm{D}\varphi$ is the Jacobian in the classical sense. If the weak divergence $\operatorname{div} A$ exists in $L^1_{\mathrm{loc}}(\Omega; \mathbb{R}^d)$, then

$$\operatorname{div} T_A = T_{\operatorname{div} A} \qquad \text{in } \mathcal{D}'(\Omega; \mathbb{R}^d).$$

Proof. For i), let $\varphi = (\varphi_1, \dots, \varphi_d)^\top \in \mathcal{D}(\Omega; \mathbb{R}^d)$. By the componentwise definition of the pairing and of the distributional gradient, we obtain

$$\langle \nabla T_u,\, \varphi\rangle = \sum_{k=1}^{d} \langle \mathrm{D}^{e_k} T_u,\, \varphi_k\rangle.$$

Using the definition of the distributional derivative, this yields

$$\langle \nabla T_u,\, \varphi\rangle = -\sum_{k=1}^{d} \langle T_u,\, \mathrm{D}^{e_k}\varphi_k\rangle = -\left\langle T_u,\, \sum_{k=1}^{d} \mathrm{D}^{e_k}\varphi_k \right\rangle = -\langle T_u,\, \operatorname{div}\varphi\rangle.$$

Since T_u is the regular distribution induced by u, we further get

$$\langle \nabla T_u,\, \varphi\rangle = -\int_\Omega u \operatorname{div}\varphi \,\mathrm{d}x.$$

Now assume that the weak gradient ∇u exists in $L^1_{\mathrm{loc}}(\Omega;\mathbb{R}^d)$. Then each weak derivative $\mathrm{D}^{e_k}u$ exists in $L^1_{\mathrm{loc}}(\Omega)$, and by the definition of weak derivative we have

$$\langle \mathrm{D}^{e_k}T_u,\,\psi\rangle = \langle T_{\mathrm{D}^{e_k}u},\,\psi\rangle \qquad \forall\,\psi\in\mathcal{D}(\Omega),\quad 1\le k\le d.$$

Hence

$$\nabla T_u = (\mathrm{D}^{e_1}T_u,\dots,\mathrm{D}^{e_d}T_u)^\top = (T_{\mathrm{D}^{e_1}u},\dots,T_{\mathrm{D}^{e_d}u})^\top = T_{\nabla u} \qquad \text{in } \mathcal{D}'(\Omega;\mathbb{R}^d).$$

For ii), let $\varphi\in\mathcal{D}(\Omega)$. By the definition of the distributional divergence and the distributional derivative,

$$\langle \operatorname{div} T_F,\,\varphi\rangle = \sum_{k=1}^d \langle \mathrm{D}^{e_k}(T_F)_k,\,\varphi\rangle = -\sum_{k=1}^d \langle (T_F)_k,\,\mathrm{D}^{e_k}\varphi\rangle.$$

By the componentwise pairing, this is

$$\langle \operatorname{div} T_F,\,\varphi\rangle = -\langle T_F,\,\nabla\varphi\rangle.$$

Since T_F is the regular distribution induced by F, we obtain

$$\langle \operatorname{div} T_F,\,\varphi\rangle = -\int_\Omega F\cdot\nabla\varphi\;\mathrm{d}x.$$

If, in addition, the weak divergence $\operatorname{div} F$ exists in $L^1_{\mathrm{loc}}(\Omega)$, then by the definition of weak divergence

$$\int_\Omega (\operatorname{div} F)\,\varphi\;\mathrm{d}x = -\int_\Omega F\cdot\nabla\varphi\;\mathrm{d}x \qquad \forall\,\varphi\in\mathcal{D}(\Omega).$$

Together with the previous identity, this implies

$$\langle \operatorname{div} T_F,\,\varphi\rangle = \langle T_{\operatorname{div} F},\,\varphi\rangle \qquad \forall\,\varphi\in\mathcal{D}(\Omega),$$

and therefore

$$\operatorname{div} T_F = T_{\operatorname{div} F} \qquad \text{in } \mathcal{D}'(\Omega).$$

For iii), let $\varphi=(\varphi_1,\dots,\varphi_d)^\top\in\mathcal{D}(\Omega;\mathbb{R}^d)$. By the componentwise definition of the pairing and of the distributional divergence of matrix-valued distributions,

$$\langle \operatorname{div} T_A,\,\varphi\rangle = \sum_{j=1}^d \left\langle \sum_{k=1}^d \mathrm{D}^{e_k}(T_A)_{jk},\,\varphi_j\right\rangle.$$

Using the definition of the distributional derivative, we get

$$\langle \operatorname{div} T_A,\,\varphi\rangle = -\sum_{j=1}^d\sum_{k=1}^d \langle (T_A)_{jk},\,\mathrm{D}^{e_k}\varphi_j\rangle.$$

Since T_A is induced by A, this becomes

$$\langle \operatorname{div} T_A,\,\varphi\rangle = -\sum_{j=1}^d\sum_{k=1}^d \int_\Omega A_{jk}\,\mathrm{D}^{e_k}\varphi_j\;\mathrm{d}x = -\int_\Omega A:\mathrm{D}\varphi\;\mathrm{d}x.$$

If the weak divergence $\operatorname{div} A$ exists in $L^1_{\mathrm{loc}}(\Omega;\mathbb{R}^d)$, then by the componentwise weak definition we have

$$\int_\Omega (\operatorname{div} A,\varphi)\;\mathrm{d}x = -\int_\Omega A:\mathrm{D}\varphi\;\mathrm{d}x \qquad \forall\,\varphi\in\mathcal{D}(\Omega;\mathbb{R}^d).$$

Hence

$$\langle \operatorname{div} T_A,\,\varphi\rangle = \langle T_{\operatorname{div} A},\,\varphi\rangle \qquad \forall\,\varphi\in\mathcal{D}(\Omega;\mathbb{R}^d),$$

which means

$$\operatorname{div} T_A = T_{\operatorname{div} A} \qquad \text{in } \mathcal{D}'(\Omega;\mathbb{R}^d).$$

■

2.5 Sobolev Spaces

Before defining Sobolev spaces, we connect the previously introduced distributional framework with the L^p-spaces from Section 2.1. Since Sobolev spaces are defined by requiring distributional derivatives to belong again to some L^p-space, it is essential that L^p-functions can be interpreted as regular distributions. The following proposition shows that every L^p-function is locally integrable and therefore canonically induces a distribution.

Lemma 2.1 (Hölder's and Cauchy-Schwarz Inequality). *Let $1 < p < \infty$ and let $p' = p/(p-1)$ be the conjugate exponent. Then for all $f \in L^p(\Omega; X)$ and $g \in L^{p'}(\Omega; X)$ Hölder's inequality*

$$\left| \int_\Omega (f,\, g)_X \,\mathrm{d}x \right| \le \|f\|_{0,p,\Omega}\, \|g\|_{0,p',\Omega}$$

holds. The special case $p = p' = 2$ yields the Cauchy-Schwarz inequality

$$(f,\, g)_{0,\Omega} \le \|f\|_{0,\Omega} \|g\|_{0,\Omega} \qquad \forall\, f, g \in L^2(\Omega; X).$$

□

Proposition 2.4 (L^p-Functions are Locally Integrable). *It holds that $L^p(\Omega; X) \subset L^1_{\mathrm{loc}}(\Omega; X)$ for all $1 \le p < \infty$. In particular, every functionf $\in L^p(\Omega; X)$ induces a regular distribution $T_f \in \mathcal{D}'(\Omega; X)$ and admits distributional derivatives of arbitrary order.*

Proof. Let $f \in L^p(\Omega; X)$ and let $K \subset \Omega$ be compact. The case $p = 1$ is trivial. Let now $p > 1$ and let $p' = p/(p-1)$ be the conjugate exponent. Since K has finite Lebesgue measure, Hölder's inequality yields

$$\int_K |f| \,\mathrm{d}x \le \|f\|_{0,p,\Omega}\, \|1\|_{0,p',\Omega} = \|f\|_{0,p,K}\, |K|^{1-1/p} \le \|f\|_{0,p,\Omega}\, |K|^{1-1/p} < \infty.$$

Hence $f|_K \in L^1(K; X)$. Since $K \subset \Omega$ was arbitrary compact, we conclude that $f \in L^1_{\mathrm{loc}}(\Omega; X)$. The remaining statement follows from the embedding $L^1_{\mathrm{loc}}(\Omega; X) \hookrightarrow \mathcal{D}'(\Omega; X)$ via $f \mapsto T_f$ and the fact that $\mathrm{D}^\alpha T_f$ is defined for every $\alpha \in \mathbb{N}_0^d$. ■

By Proposition 2.4, every $f \in L^p(\Omega; X)$ can be identified with the regular distribution $T_f \in \mathcal{D}'(\Omega; X)$ given by

$$\langle T_f,\, \varphi \rangle = \int_\Omega (f,\, \varphi)_X \,\mathrm{d}x, \qquad \forall\, \varphi \in \mathcal{D}(\Omega; X).$$

In the following, we will use this identification implicitly whenever no confusion can arise. In the Hilbert case $p = 2$, this pairing is consistent with the L^2-inner product introduced in Proposition 2.1. Indeed, since $\varphi \in \mathcal{D}(\Omega; X)$ has compact support, we have $\varphi \in L^2(\Omega; X)$, and therefore

$$\langle T_f,\, \varphi \rangle = \int_\Omega (f,\, \varphi)_X \,\mathrm{d}x = (f,\, \varphi)_{0,\Omega}, \qquad f \in L^2(\Omega; X),\ \varphi \in \mathcal{D}(\Omega; X).$$

Thus, on test functions, the duality pairing of the regular distribution T_f coincides with the L^2-inner product. In the Sobolev setting, we therefore use the terms distributional derivative and weak derivative interchangeably whenever the derivative is represented by an L^p-function. We continue with the definition of Sobolev spaces of integer order. For now, we only need the case $X = \mathbb{R}^m$, $m \ge 1$

Definition 2.13 (Sobolev Spaces of Integer Order)**.** *Let $m \geq 1$, $k \in \mathbb{N}_0$ and $1 \leq p < \infty$. The* `Sobolev space` *$W^{k,p}(\Omega;\mathbb{R}^m)$ is defined by*

$$W^{k,p}(\Omega;\mathbb{R}^m) := \left\{ u \in L^p(\Omega;\mathbb{R}^m) \;\middle|\; \mathrm{D}^\alpha u \in L^p(\Omega;\mathbb{R}^m) \;\forall\, \alpha \in \mathbb{N}_0^d,\ |\alpha| \leq k \right\},$$

where $\mathrm{D}^\alpha u$ is understood in the weak sense. If $m = 1$, we abbreviate $W^{k,p}(\Omega) := W^{k,p}(\Omega;\mathbb{R}^1)$ and we adopt the usual conventions

$$H^k(\Omega;\mathbb{R}^m) := W^{k,2}(\Omega;\mathbb{R}^m) \quad \text{and} \quad W^{0,p}(\Omega;\mathbb{R}^m) := L^p(\Omega;\mathbb{R}^m).$$

Moreover, we equip $W^{k,p}(\Omega;\mathbb{R}^m)$ with the norm

$$\|u\|_{k,p,\Omega} := \left(\sum_{|\alpha| \leq k} \|\mathrm{D}^\alpha u\|_{0,p,\Omega}^p \right)^{1/p}, \qquad u \in W^{k,p}(\Omega;\mathbb{R}^m).$$

For $k \geq 1$, we also introduce the semi-norm

$$|u|_{k,p,\Omega} := \left(\sum_{|\alpha| = k} \|\mathrm{D}^\alpha u\|_{0,p,\Omega}^p \right)^{1/p}, \qquad u \in W^{k,p}(\Omega;\mathbb{R}^m).$$

Again, the special case $p = 2$ provides us with a Hilbert space and we introduce the following notation for the corresponding inner product.

Proposition 2.5 ($H^k(\Omega;\mathbb{R}^m)$ is a Hilbert space)**.** *The Sobolev space $H^k(\Omega;\mathbb{R}^m)$ together with the inner product*

$$(u,\, v)_{k,\Omega} := \sum_{|\alpha| \leq k} (\mathrm{D}^\alpha u,\, \mathrm{D}^\alpha v)_{0,\Omega}, \qquad u, v \in H^k(\Omega;\mathbb{R}^m)$$

is a Hilbert space and it induces the norm

$$\|u\|_{k,\Omega} = \left(\sum_{|\alpha| \leq k} \|\mathrm{D}^\alpha u\|_{0,\Omega}^2 \right)^{1/2} < \infty, \qquad u \in H^k(\Omega;\mathbb{R}^m).$$

□

After the integer-order spaces $W^{k,p}(\Omega;\mathbb{R}^m)$, we also need Sobolev spaces of non-integer order. The main reason is that trace operators naturally lead to boundary spaces of fractional order. In particular, traces of $H^1(\Omega;\mathbb{R}^m)$-functions are, in general, not measured in another integer-order Sobolev space, but in a space of order $1/2$ on the boundary and similarly on boundary parts $\Gamma_0 \subset \Gamma$. For this reason, it is convenient to introduce fractional Sobolev spaces already at this stage. For integer orders, regularity is encoded by requiring weak derivatives $\mathrm{D}^\alpha u$ to belong to L^p. For a non-integer order $s = k + \theta$ with $\theta \in (0,1)$, the remaining fractional part θ is measured by difference quotients. More precisely, one measures how strongly a function oscillates between pairs of points $x, y \in \Omega$, with a weight that becomes singular as $|x - y| \to 0$. This leads to the following quantity.

Definition 2.14 (Slobodeckij Seminorm)**.** *Let $m \geq 1$, $1 \leq p < \infty$, and $\theta \in (0,1)$. For a* *measurable function $u : \Omega \to \mathbb{R}^m$, we define the* `Slobodeckij semi-norm`

$$|u|_{\theta,p,\Omega} := \left(\int_\Omega \int_\Omega \frac{|u(x) - u(y)|^p}{|x - y|^{d+p\theta}} \, \mathrm{d}x \, \mathrm{d}y \right)^{1/p},$$

whenever the right-hand side is finite.

The quantity $|u|_{\theta,p,\Omega}$ is only a semi-norm, since it vanishes on constant functions. Combined with an L^p-term, however, it yields a genuine norm.

Definition 2.15 (Sobolev Spaces of Fractional Order)**.** *Let $m \geq 1$, $1 \leq p < \infty$, and let $s = k + \theta$ with $k \in \mathbb{N}_0$ and $\theta \in (0, 1)$. The* `Sobolev space` *of fractional order s is defined by*

$$W^{s,p}(\Omega; \mathbb{R}^m) := \left\{ u \in W^{k,p}(\Omega; \mathbb{R}^m) \;\middle|\; |\mathrm{D}^\alpha u|_{\theta,p,\Omega} < \infty \;\forall\, \alpha \in \mathbb{N}_0^d,\; |\alpha| = k \right\},$$

where $\mathrm{D}^\alpha u$ is understood in the weak sense. If $m = 1$, we abbreviate $W^{s,p}(\Omega) := W^{s,p}(\Omega; \mathbb{R}^1)$ and we adopt the usual convention

$$H^s(\Omega; \mathbb{R}^m) := W^{s,2}(\Omega; \mathbb{R}^m).$$

Moreover, we define the fractional Sobolev semi-norm

$$|u|_{s,p,\Omega} := \left(\sum_{|\alpha|=k} |\mathrm{D}^\alpha u|^p_{\theta,p,\Omega} \right)^{1/p}, \qquad u \in W^{s,p}(\Omega; \mathbb{R}^m),$$

and equip $W^{s,p}(\Omega; \mathbb{R}^m)$ with the norm

$$\|u\|_{s,p,\Omega} := \left(\|u\|^p_{k,p,\Omega} + |u|^p_{s,p,\Omega} \right)^{1/p}, \qquad u \in W^{s,p}(\Omega; \mathbb{R}^m).$$

For general non-integer order $s = k + \theta$, one applies this semi-norm to all weak derivatives of order k. As before, the Hilbert case $p = 2$ is of particular importance for the subsequent variational analysis. In particular, the case $s = 1/2$ will later arise naturally in connection with trace operators and boundary spaces.

Proposition 2.6 ($H^s(\Omega; \mathbb{R}^m)$ is a Hilbert Space)**.** *Let $m \geq 1$ and let $s = k + \theta$ with $k \in \mathbb{N}_0$ and $\theta \in (0, 1)$. Then $H^s(\Omega; \mathbb{R}^m)$ is a Hilbert space with inner product*

$$(u,\, v)_{s,\Omega} := (u,\, v)_{k,\Omega} + \sum_{|\alpha|=k} \int_\Omega \int_\Omega \frac{(\mathrm{D}^\alpha u(x) - \mathrm{D}^\alpha u(y)) \cdot (\mathrm{D}^\alpha v(x) - \mathrm{D}^\alpha v(y))}{|x - y|^{d+2\theta}} \, \mathrm{d}x \, \mathrm{d}y$$

for $u, v \in H^s(\Omega; \mathbb{R}^m)$. The induced norm is

$$\|u\|_{s,\Omega} = \left(\|u\|^2_{k,\Omega} + \sum_{|\alpha|=k} |\mathrm{D}^\alpha u|^2_{\theta,2,\Omega} \right)^{1/2}, \qquad u \in H^s(\Omega; \mathbb{R}^m). \tag{2.1}$$

2.6 Trace Operators & Boundary Spaces

In variational formulations, boundary conditions and boundary forces are imposed on traces of bulk functions. For functions in $H^1(\Omega; \mathbb{R}^m)$, pointwise boundary values are in general not defined. The trace theorem (see [1, Theorem 2.6]) provides a canonical replacement by a continuous linear trace operator taking values in a fractional-order Sobolev space on the boundary. Throughout this subsection, we assume that $\Omega \subset \mathbb{R}^d$ is a bounded Lipschitz domain and write $\Gamma := \partial\Omega$.

Theorem 2.1 (Trace Theorem). *Let $m \geq 1$. There exists a unique linear continuous operator*

$$\gamma : H^1(\Omega; \mathbb{R}^m) \to H^{1/2}(\Gamma; \mathbb{R}^m)$$

which coincides with the classical restriction on sufficiently regular functions. Moreover, there exists a constant $C_{\mathrm{tr}} > 0$ such that

$$\|\gamma u\|_{\frac{1}{2},\Gamma} \leq C_{\mathrm{tr}} \|u\|_{1,\Omega} \qquad \forall\, u \in H^1(\Omega; \mathbb{R}^m).$$

Furthermore, γ is surjective and admits a linear continuous right inverse

$$\mathcal{L} : H^{1/2}(\Gamma; \mathbb{R}^m) \to H^1(\Omega; \mathbb{R}^m)$$

such that

$$\gamma(\mathcal{L}g) = g \qquad \forall\, g \in H^{1/2}(\Gamma; \mathbb{R}^m).$$

The operator $\mathcal{L}$ is called the lifting operator.

The trace theorem motivates the following definition of boundary spaces.

Definition 2.16 (Boundary Sobolev Space of Order 1/2). *We define equip the space*

$$H^{1/2}(\Gamma; \mathbb{R}^m) = \gamma(H^1(\Omega; \mathbb{R}^m)).$$

with the quotient norm

$$\|g\|_{\frac{1}{2},\Gamma} := \inf \left\{ \|u\|_{1,\Omega} \;\middle|\; u \in H^1(\Omega; \mathbb{R}^m),\ \gamma u = g \right\}, \qquad g \in H^{1/2}(\Gamma; \mathbb{R}^d) \tag{2.2}$$

Notice, how the notation of (2.2) and (2.1) coincide, which is due to equivalency of both of these norms on $H^{1/2}(\Gamma; \mathbb{R}^m)$ (see [1, Proposition 2.14]). Theorem 2.1 therefore holds for both of these norms and if used later on, we will opt for the second option.

Definition 2.17 (Dual Boundary Space). *The dual boundary space of order $-1/2$ is defined by*

$$H^{-1/2}(\Gamma; \mathbb{R}^m) := (H^{1/2}(\Gamma; \mathbb{R}^m))'.$$

For $\lambda \in H^{-1/2}(\Gamma; \mathbb{R}^m)$ and $g \in H^{1/2}(\Gamma; \mathbb{R}^m)$, we denote the duality pairing by $\langle \lambda, g \rangle_\Gamma$.

In applications, boundary conditions are often prescribed only on a measurable boundary part $\Gamma_0 \subset \Gamma$. We therefore also use trace spaces on boundary parts.

Definition 2.18 (Trace Spaces on Boundary Parts). *Let $\Gamma_0 \subset \Gamma$ be measurable. We define*

$$H^{1/2}(\Gamma_0; \mathbb{R}^m) := \left\{ g : \Gamma_0 \to \mathbb{R}^m \;\middle|\; \exists\, \widetilde{g} \in H^{1/2}(\Gamma; \mathbb{R}^m) \text{ with } \widetilde{g}|_{\Gamma_0} = g \right\},$$

with norm

$$\|g\|_{\frac{1}{2},\Gamma_0} := \inf \left\{ \|\widetilde{g}\|_{\frac{1}{2},\Gamma} \;\middle|\; \widetilde{g} \in H^{1/2}(\Gamma; \mathbb{R}^m),\ \widetilde{g}|_{\Gamma_0} = g \right\}.$$

If $m = 1$, we abbreviate $H^{1/2}(\Gamma_0) := H^{1/2}(\Gamma_0; \mathbb{R}^1)$.

Definition 2.19 (Dual Boundary Space on Boundary Parts)**.** *Let* $\Gamma_0 \subset \Gamma$ *be measurable. We define*

$$H^{-1/2}(\Gamma_0;\mathbb{R}^m) := \big(H^{1/2}(\Gamma_0;\mathbb{R}^m)\big)',$$

and denote the duality pairing by

$$\langle \lambda,\, g\rangle_{\Gamma_0}, \qquad \lambda \in H^{-1/2}(\Gamma_0;\mathbb{R}^m),\ g \in H^{1/2}(\Gamma_0;\mathbb{R}^m).$$

For later use in Signorini-type boundary conditions, we note that for $u \in H^1(\Omega;\mathbb{R}^d)$ the trace γu belongs to $H^{1/2}(\Gamma;\mathbb{R}^d)$ and can therefore be restricted to boundary parts such as a contact boundary $\Gamma_C \subset \Gamma$ or a Dirichlet boundary $\Gamma_D \subset \Gamma$. Boundary data will be measured in the corresponding spaces $H^{1/2}(\Gamma_0;\mathbb{R}^d)$, while boundary functionals act naturally in the dual spaces $H^{-1/2}(\Gamma_0;\mathbb{R}^d)$.

3 Signorini's Problem

3.1 Physical and Geometric Setup of the Problem

We consider an elastic body occupying a reference configuration $\Omega \subset \mathbb{R}^d$, $d \in \{2,3\}$. Under the action of external forces and possible contact with a rigid foundation, the body deforms. In the *small–strain* framework, the deformation is described by the *displacement field* $u : \Omega \to \mathbb{R}^d$, and a material point $x \in \Omega$ is mapped to its deformed position

$$\varphi(x) := x' := x + u(x), \qquad x \in \overline{\Omega}.$$

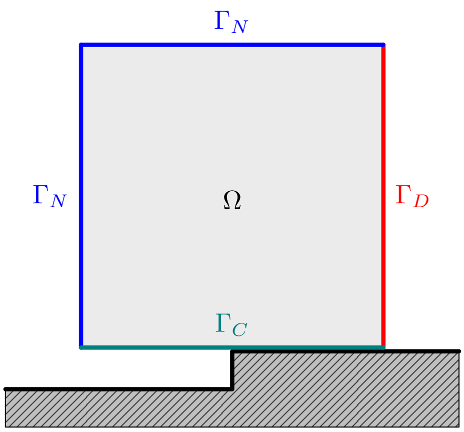
The aim of Signorini's problem is to determine an *equilibrium* displacement u under standard boundary conditions and *unilateral* contact constraints on the part of the boundary that may touch the rigid foundation. Throughout the whole report, $\Omega \subset \mathbb{R}^d$, $d \in \{2,3\}$ denotes an open, bounded and connected *Lipschitz* domain. The later means, that the boundary $\Gamma := \partial\Omega$ can locally be described as the graph of a Lipschitz continuous function. This ensures for example, that the outward normal vector $\mathrm{n}(x)$ is well–defined almost everywhere. The boundary is decomposed into three pairwise disjoint parts

$$\Gamma = \overline{\Gamma_D} \,\dot\cup\, \overline{\Gamma_N} \,\dot\cup\, \overline{\Gamma_C}$$

which are assumed to be open in Γ and where each part is subject to different boundary conditions to model different physical interactions (See Figure 1). On Γ_D, we apply *homogeneous Dirichlet* boundary conditions. This forces the body to be *clamped* and this part, i.e. the displacement should not act on Γ_D,

$$\forall x \in \Gamma_D : \; x + u(x) = x \qquad \Longrightarrow \qquad (\gamma u)|_{\Gamma_D} = 0.$$

The part Γ_N may be subject to a *Neumann* boundary condition, which means we impose a known *surface force density*. Γ_C denotes the *potential contact* boundary. We assume that Γ_C and Γ_D have positive $(d-1)$–Lebesgue measure. The rigid foundation is positioned so that any actual contact, both initially and in the deformed equilibrium configuration, may only occur on Γ_C. While Γ_C is prescribed a priori, the *active contact zone* $\Gamma_{\text{act}}(u) \subset \Gamma_C$ is unknown and depends on the solution u. Later on, the boundary conditions imposed on Γ_C enforce *unilateral* contact through *nonpenetration* and *complementarity*.



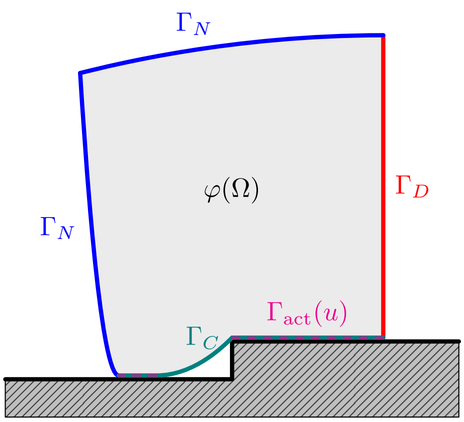


Figure 1: 2D example of Signorini's problem before (left) and after displacement (right).

3.2 Weak Formulation

3.2.1 Admissible Displacements

We start by introducing the energy space and the admissible set of displacements. Homogeneous Dirichlet boundary conditions on Γ_D are incorporated by

$$V := \left\{ v \in H^1(\Omega; \mathbb{R}^d) \,\middle|\, \gamma v = 0 \text{ on } \Gamma_D \right\} \subset H^1(\Omega; \mathbb{R}^d). \tag{3.1}$$

For $v \in V$ we define the normal component of the boundary displacement on Γ_C by

$$v_{\mathrm{n}} := (\gamma v) \cdot \mathrm{n} \qquad \text{on } \Gamma_C, \tag{3.2}$$

where n is the outward unit normal on Γ. The unilateral nature of contact is expressed by the *nonpenetration* constraint. The rigid foundation is located on the exterior side of Γ_C in direction of n. Hence the body is not allowed to move across Γ_C into the obstacle. In the small–strain setting this yields

$$u_{\mathrm{n}} \leq 0 \qquad \text{a.e. on } \Gamma_C.$$

This motivates the nonempty, closed and convex cone ([1, Prop. 3.1]) of admissible displacements

$$K := \{ v \in V \mid v_{\mathrm{n}} \leq 0 \text{ a.e. on } \Gamma_C \} \subset V. \tag{3.3}$$

3.2.2 Small Strain Elasticity

We start by defining the *small strain tensor* for a displacement function $u \in H^1(\Omega; \mathbb{R}^d)$, which is similar to the symmetric part of square matrices but for the gradient ∇u of u and through which the deformation of the body $\overline{\Omega}$ by u is quantified.

Definition 3.1 (Small Strain Tensor)**.** *The* `(small) strain tensor` ε *is defined by*

$$\varepsilon : H^1(\Omega; \mathbb{R}^d) \to L^2(\Omega; \mathrm{Sym}^2(\mathbb{R}^d)), \qquad u \mapsto \varepsilon(u) := \frac{1}{2}\left(\nabla u + \nabla u^\top \right).$$

In linear elasticity, stresses are assumed to depend linearly on strains. The material response is encoded in the fourth order *elasticity tensor* $\mathcal{A}$, which maps a symmetric strain tensor to the corresponding *Cauchy stress tensor*. The tensor field $\mathcal{A}$ is prescribed by the material and may vary in space, describing a *heterogeneous* body.

Definition 3.2 (Elasticity tensor field)**.** *An* `elasticity tensor field` *is a fourth order tensor*

$$\mathcal{A} : \Omega \to \mathcal{L}(\mathrm{Sym}^2(\mathbb{R}^d); \mathrm{Sym}^2(\mathbb{R}^d))$$

such that the coefficients satisfy the symmetries

$$\mathcal{A}_{ijkl}(x) = \mathcal{A}_{jikl}(x) = \mathcal{A}_{ijlk}(x) = \mathcal{A}_{klij}(x), \qquad \forall\, 1 \leq i,j,k,l \leq d,$$

and the uniform ellipticity condition, i.e. there exists $\alpha_{\mathcal{A}} > 0$ *such that for a.e.* $x \in \Omega$

$$s : \mathcal{A}(x) : s \geq \alpha_{\mathcal{A}} \, (s : s) \qquad \forall\, s \in \mathrm{Sym}^2(\mathbb{R}^d).$$

Moreover, there exists $C_{\mathcal{A}} > 0$ *such that for a.e.* $x \in \Omega$

$$|\mathcal{A}_{ijkl}(x)| \leq C_{\mathcal{A}} \qquad \forall\, 1 \leq i,j,k,l \leq d.$$

The *internal* and *surface forces* in the body $\overline{\Omega}$ are described by the *Cauchy stress tensor*.

Definition 3.3 (Cauchy stress tensor)**.** *Let $\mathcal{A}$ be an elasticity tensor field. We define the* (linear) Cauchy stress tensor *by given a.e. by*

$$\sigma : H^1(\Omega; \mathbb{R}^d) \to L^2(\Omega; \mathrm{Sym}^2(\mathbb{R}^d)), \qquad u \mapsto \mathcal{A} : \varepsilon(u).$$

On V, we define the symmetric bilinear form

$$a(u, v) := \int_\Omega \sigma(u) : \varepsilon(v) \, \mathrm{d}x, \qquad u, v \in V \tag{3.4}$$

representing the *virtual work of internal forces* and the linear bounded functional

$$\ell(v) := \int_\Omega f \cdot v \, \mathrm{d}x + \int_{\Gamma_N} F \cdot \gamma v \, \mathrm{d}s, \qquad v \in V, \tag{3.5}$$

with *volume force density* $f \in L^2(\Omega; \mathbb{R}^d)$ on the body and *surface loads* $F \in L^2(\Gamma_N; \mathbb{R}^d)$ on the Neumann boundary, representing the *virtual work of external forces*. In the following, we want to prove standard properties of a and ℓ. For this, we will need the following inequalities. The first inequality is a result of *Korn's second inequality* ([1, Theorem 3.3] and [1, Section 3.5.2]) and the *Peetre–Tartar* Lemma ([1, Lemma 3.5]). Therefore only the proof of the second one is given.

Lemma 3.1. *Assuming $|\Gamma_D| > 0$, there exists a constant $C_K > 0$ such that*

$$C_K^{-1} \|v\|_{1,\Omega} \leq \|\varepsilon(v)\|_{0,\Omega} \leq \|v\|_{1,\Omega} \qquad \forall\, v \in V.$$

Proof. We prove the second inequality. For all $v \in V$ we have

$$|\varepsilon(v)(x)| = \left| \frac{1}{2}(\nabla v(x) + \nabla v^\top) \right| \leq \frac{1}{2}(|\nabla v(x)| + |\nabla v(x)^\top|) = |\nabla v(x)|$$

and thus

$$\|\varepsilon(v)\|_{0,\Omega}^2 = \int_\Omega |\varepsilon(v)(x)|^2 \, \mathrm{d}x \leq \int_\Omega |\nabla v(x)|^2 \, \mathrm{d}x = \|\nabla v\|_{0,\Omega}^2 \leq \|v\|_{1,\Omega}^2,$$

where the last inequality holds due to

$$\|v\|_{1,\Omega}^2 = \|v\|_{0,\Omega}^2 + \|\nabla v\|_{0,\Omega}^2.$$

■

Proposition 3.1. *The symmetric bilinear form $a : V \times V \to \mathbb{R}$ defined in (3.4) is*

i) bounded, *i.e. for all $v, w \in V$ there exists $C_1 < \infty$ such that*

$$|a(u, v)| \leq C_1 \|u\|_{1,\Omega} \|v\|_{1,\Omega} \qquad \textit{with} \qquad C_1 = C_{\mathcal{A}},$$

ii) V-elliptic, *i.e. for all $v \in V \setminus \{0\}$ there exist $C_2 > 0$ such that*

$$a(v, v) \geq C_2 \|v\|_{1,\Omega}^2 \qquad \textit{with} \qquad C_2 = \frac{\alpha_{\mathcal{A}}}{C_K^2}.$$

Moreover, the linear functional ℓ defined in (3.5) is bounded, *i.e. for all $v \in V$ there exists $C_3 > 0$ such that*

$$|\ell(v)| \leq C_3 \|v\|_{1,\Omega} \qquad \textit{with} \qquad C_3 = \max\{1, C_{\mathrm{tr}}\} \left(\|f\|_{0,\Omega} + \|F\|_{0,\Gamma_N}\right) \|v\|_{1,\Omega}. \tag{3.6}$$

Proof. Let $v, w \in V$ be arbitrary. Then

$$|a(v,w)| \leq \|\mathcal{A} : \varepsilon(v)\|_{0,\Omega} \, \|\varepsilon(w)\|_{0,\Omega}$$

by the Cauchy–Schwarz inequality. With Definition 3.2 there exists $C_{\mathcal{A}} > 0$ such that

$$\|\mathcal{A} : \varepsilon(v)\|_{0,\Omega} \leq C_{\mathcal{A}} \, \|\varepsilon(v)\|_{0,\Omega}.$$

Combining this with the second inequality form Lemma 3.1 we have

$$|a(v,w)| \leq C_{\mathcal{A}} \, \|\varepsilon(v)\|_{0,\Omega} \, \|\varepsilon(w)\|_{0,\Omega} \leq C_{\mathcal{A}} \, \|v\|_{1,\Omega} \, \|w\|_{1,\Omega}.$$

The ellipticity of a follows by

$$a(v,v) = \int_{\Omega} \varepsilon(v) : \mathcal{A} : \varepsilon(v) \, \mathrm{d}x \geq \alpha_{\mathcal{A}} \int_{\Omega} \varepsilon(v) : \varepsilon(v) = \alpha_{\mathcal{A}} \, \|\varepsilon(v)\|_{0,\Omega}^2,$$

where $\alpha_{\mathcal{A}} > 0$ exists due to Definition 3.2. The first inequality of Lemma 3.1 finally yields

$$a(v,v) \geq \alpha_{\mathcal{A}} \, \|\varepsilon(v)\|_{0,\Omega}^2 \geq \alpha_{\mathcal{A}} \left(C_K^{-1} \, \|v\|_{1,\Omega} \right)^2 = \frac{\alpha_{\mathcal{A}}}{C_K^2} \, \|v\|_{1,\Omega}^2.$$

Again by Cauchy–Schwarz inequality,

$$|\ell(v)| \leq \|f\|_{0,\Omega} \, \|v\|_{0,\Omega} + \|F\|_{0,\Gamma_N} \, \|\gamma v\|_{0,\Gamma_N} \qquad \forall \, v \in V.$$

By the Trace Theorem 2.1 there exists $C_{\mathrm{tr}} > 0$ such that

$$\|\gamma v\|_{0,\Gamma_N} \leq \|\gamma v\|_{0,\Gamma} \leq \|\gamma v\|_{\frac{1}{2},\Gamma} \leq C_{\mathrm{tr}} \, \|v\|_{1,\Omega} \qquad \forall \, v \in V.$$

Finally,

$$|\ell(v)| \leq (\|f\|_{0,\Omega} + C_{\mathrm{tr}} \, \|F\|_{0,\Gamma_N}) \, \|v\|_{1,\Omega} \leq \max\{1, C_{\mathrm{tr}}\} \, (\|f\|_{0,\Omega} + \|F\|_{0,\Gamma_N}) \, \|v\|_{1,\Omega} \qquad \forall \, v \in V.$$

∎

3.2.3 Energy Minimization and Well-posedness

The total potential energy of the elastic body is given by the quadratic functional

$$\mathcal{J}(v) := \frac{1}{2} a(v,v) - \ell(v), \qquad v \in V. \tag{3.7}$$

In the presence of unilateral contact, the admissible displacements are restricted to the closed convex set $K \subset V$. We therefore consider the following constrained minimization problem.

Problem 3.1 (Signorini's Problem – Energy Minimization)**.** *Find $u \in K$ such that*

$$\mathcal{J}(u) = \min_{v \in K} \mathcal{J}(v) = \min_{v \in K} \frac{1}{2} a(v,v) - \ell(v).$$

We now what to prove well-posedness of the above Problem 3.1. This consist of existence and uniqueness of a solution and continuous dependency of the solution on the load functional ℓ. We start by the uniqueness, which follows directly by the strict convexity of $\mathcal{J}$ and the convexity of K.

Lemma 3.2 (Strict Convexity of $\mathcal{J}$)**.** *The Functional $\mathcal{J}$ defined in (3.7) is strictly convex.*

Proof. For $u, v \in V$ and $t \in [0, 1]$, we set $w_t := (1-t)u + tv$. Then

$$a(w_t, w_t) = (1-t)a(u,u) + ta(v,v) - t(1-t)a(u-v, u-v)$$

and

$$\ell(w_t) = (1-t)\ell(u) + t\ell(v).$$

Together, we have

$$\mathcal{J}(w_t) = (1-t)\mathcal{J}(u) + t\mathcal{J}(v) - \frac{1}{2}t(1-t)a(u-v, u-v).$$

If $u \neq v$ then $u - v \neq 0$ and thus by V-ellipticity of a we have $a(u-v, u-v) > 0$ and $t(1-t) > 0$. This implies

$$\mathcal{J}(w_t) < (1-t)\mathcal{J}(u) + t\mathcal{J}(v).$$

■

Proposition 3.2 (Uniqueness of Minimizer of $\mathcal{J}$)**.** *If there exists a minimizer of the functional $\mathcal{J}$ defined in (3.7), then it is unique.*

Proof. Assume that there exist two minimizers $u_1, u_2 \in K$ of $\mathcal{J}$ on K, i.e.

$$\mathcal{J}(u_1) = \min_{v \in K} \mathcal{J}(v) = \mathcal{J}(u_2). \tag{3.8}$$

If $u_1 = u_2$ there is nothing to show. Hence assume $u_1 \neq u_2$. Since K is convex, for every $t \in (0, 1)$ the convex combination

$$w_t := (1-t)u_1 + tu_2$$

belongs to K. By Lemma 3.2, $\mathcal{J}$ is strictly convex and therefore we have

$$\mathcal{J}(w_t) < (1-t)\mathcal{J}(u_1) + t\mathcal{J}(u_2) \qquad \forall\, t \in (0, 1).$$

Using (3.8), this yields

$$\mathcal{J}(w_t) < (1-t)\min_{v \in K} \mathcal{J}(v) + t \min_{v \in K} \mathcal{J}(v) = \min_{v \in K} \mathcal{J}(v).$$

This contradicts the definition of the minimum, since $w_t \in K$. Therefore the assumption $u_1 \neq u_2$ is false and we conclude $u_1 = u_2$. ■

It remains to show the existence of at least one minimizer. For this, we verify *coercivity* and *weakly lower semicontinuity* of the functional $\mathcal{J}$. The proof requires the following lemma.

Lemma 3.3 (Weak lower semicontinuity of the norm)**.** *Let $(X, \|\cdot\|_X)$ be a normed space. Then $\|\cdot\|_X : X \to \mathbb{R}_{\geq 0}$ is a weakly lower semicontinuous functional, i.e. if $x_j \rightharpoonup x$ in X, then*

$$\|x\|_X \leq \liminf_{j \to \infty} \|x_j\|_X.$$

Proof. Let $J : X \to X^{**}$ be the canonical embedding defined by

$$(Jx)(\varphi) := \varphi(x) \qquad \forall\, \varphi \in X^*.$$

By the Hahn–Banach theorem, J is an isometry, i.e. $\|Jx\|_{X^{**}} = \|x\|_X$ for all $x \in X$. Therefore,

$$\|x\|_X = \|Jx\|_{X^{**}} = \sup_{\substack{\varphi \in X^* \\ \|\varphi\|_{X^*} \leq 1}} |(Jx)(\varphi)| = \sup_{\substack{\varphi \in X^* \\ \|\varphi\|_{X^*} \leq 1}} |\varphi(x)|.$$

If $x_j \rightharpoonup x$ in X, then $\varphi(x_j) \to \varphi(x)$ for all $\varphi \in X^*$ and hence

$$\|x\|_X = \sup_{\substack{\varphi \in X^* \\ \|\varphi\|_{X^*} \leq 1}} |\varphi(x)| \leq \liminf_{j \to \infty} \sup_{\substack{\varphi \in X^* \\ \|\varphi\|_{X^*} \leq 1}} |\varphi(x_j)| = \liminf_{j \to \infty} \|x_j\|_X.$$

■

Lemma 3.4 ($\mathcal{J}$ is coercive and w.l.s.c.). *The functional* $\mathcal{J}$ *defined in* (3.7) *is*

i) coercive *on* V, *i.e.*

$$\mathcal{J}(v) \to \infty \qquad \text{as} \qquad \|v\|_{1,\Omega} \to \infty,$$

ii) weakly lower semicontinuous *on* V, *i.e. if* $v_j \rightharpoonup v$ *in* V, *then*

$$\mathcal{J}(v) \leq \liminf_{j\to\infty} \mathcal{J}(v_j).$$

Proof. By boundedness of ℓ we have $|\ell(v)| \leq \|\ell\|_{-1,\Omega}\|v\|_{1,\Omega}$. Hence

$$\mathcal{J}(v) = \frac{1}{2}\, a(v,v) - \ell(v) \geq \frac{\alpha}{2}\, \|v\|_{1,\Omega}^2 - \|\ell\|_{-1,\Omega}\, \|v\|_{1,\Omega} \xrightarrow{\|v\|_{1,\Omega}\to\infty} \infty.$$

For $s \in L^2(\Omega; \mathrm{Sym}^2(\mathbb{R}^d))$ set

$$\|s\|_{\mathcal{A}}^2 := \int_\Omega s : \mathcal{A} : s \, \mathrm{d}x.$$

By uniform ellipticity of $\mathcal{A}$, the map $\|\cdot\|_{\mathcal{A}}$ defines a norm on $L^2(\Omega; \mathrm{Sym}^2(\mathbb{R}^d))$ equivalent to $\|\cdot\|_{0,\Omega}$. Moreover, $a(v,v) = \|\varepsilon(v)\|_{\mathcal{A}}^2$. The mapping $v \mapsto \varepsilon(v)$ is linear and bounded from V to $L^2(\Omega; \mathrm{Sym}^2(\mathbb{R}^d))$. Hence, if $v_j \rightharpoonup v$ in V, then

$$\varepsilon(v_j) \rightharpoonup \varepsilon(v) \qquad \text{in } L^2(\Omega; \mathrm{Sym}^2(\mathbb{R}^d)).$$

Since norms are weakly lower semicontinuous by Lemma 3.3, we obtain

$$a(v,v) = \|\varepsilon(v)\|_{\mathcal{A}}^2 \leq \liminf_{j\to\infty} \|\varepsilon(v_j)\|_{\mathcal{A}}^2 = \liminf_{j\to\infty} a(v_j,v_j).$$

Furthermore, $\ell \in V'$ is weakly continuous, hence $\ell(v_j) \to \ell(v)$. Therefore,

$$\mathcal{J}(v) = \frac{1}{2}\, a(v,v) - \ell(v) \leq \frac{1}{2}\, \liminf_{j\to\infty} a(v_j,v_j) - \lim_{j\to\infty} \ell(v_j) = \liminf_{j\to\infty} \mathcal{J}(v_j).$$

∎

Proposition 3.3 (Existence of Minimizer of $\mathcal{J}$). *The functional* $\mathcal{J}$ *defined in 3.7 has at least one minimizer* $u \in K$.

Proof. Set

$$M := \inf_{v\in K} \mathcal{J}(v) \in [-\infty, \infty).$$

Since $0 \in K$ we have $M \leq \mathcal{J}(0) = 0$, hence M is finite from above. Let $\{v_j\}_{j\in\mathbb{N}} \subset K$ be a minimizing sequence, i.e. $\mathcal{J}(v_j) \to M$ as $j \to \infty$. By Lemma 3.4 *i)*, $\mathcal{J}$ is coercive and thus, the sequence $\{v_j\}_{j\in\mathbb{N}}$ is bounded in V. Since V is a Hilbert space, there exist a subsequence $\{v_{j_k}\}_{k\in\mathbb{N}}$ and some $u \in V$ such that

$$v_{j_k} \rightharpoonup u \qquad \text{in } V.$$

As $K \subset V$ is closed and convex, it is weakly closed in V. Hence $u \in K$. By Lemma 3.4 *ii)*, $\mathcal{J}$ is weakly lower semicontinuity and thus we obtain

$$\mathcal{J}(u) \leq \liminf_{k\to\infty} \mathcal{J}(v_{j_k}) = M = \inf_{v\in K} \mathcal{J}(v),$$

so u is a minimizer of $\mathcal{J}$ on K. ∎

Corollary 3.1 (Well-posedness of Problem 3.1)**.** *The functional $\mathcal{J}$ defined in 3.7 admits a unique minimizer $u \in K$. Moreover, the minimizer satisfies the a-priori bound*

$$\|u\|_{1,\Omega} \leq \frac{2\,C_K^2\,\max\{1, C_{\mathrm{tr}}\}}{\alpha_{\mathcal{A}}}\left(\|f\|_{0,\Omega} + \|F\|_{0,\Gamma_N}\right). \tag{3.9}$$

Consequently, Problem 3.1 has a unique stable solution.

Proof. By Proposition 3.3 we have existence of a solution $u \in K$ to Problem 3.1 and by Proposition 3.2 it is unique. It remains to prove the a-priori bound. Since u is the unique minimizer, it holds

$$\mathcal{J}(u) \leq \mathcal{J}(0) = 0 \qquad \Longrightarrow \qquad \frac{1}{2}\,a(u,u) \leq \ell(u).$$

By Proposition 3.1 *ii)* we have

$$a(u,u) \geq \frac{\alpha_{\mathcal{A}}}{C_K^2}\,\|u\|_{1,\Omega}^2$$

and thus

$$\frac{\alpha_{\mathcal{A}}}{2\,C_K^2}\,\|u\|_{1,\Omega}^2 \leq \frac{1}{2}\,a(u,u) \leq |\ell(u)| \leq \max\{1, C_{\mathrm{tr}}\}\left(\|f\|_{0,\Omega} + \|F\|_{0,\Gamma_N}\right)\|u\|_{1,\Omega},$$

where the last inequality is (3.6). Dividing both sides by $\|u\|_{1,\Omega}$ yields the desired estimate (3.9). For $u = 0$ the estimate is trivial. ∎

3.2.4 Variational inequality as an optimality condition

Finally, we turn to the *weak formulation* of Signorini's problem. Unlike the weak formulation of conventional simple boundary value problems, here we do not end up with a variational equation, but with a variational inequality. This is derived by a first-order optimality condition on the functional $\mathcal{J}$.

Lemma 3.5 (Fréchet differentiability of $\mathcal{J}$)**.** *The functional $\mathcal{J}$ defined in (3.7) is Fréchet differentiable on V. Its derivative at $u \in V$ is given by*

$$\mathcal{J}'(u)[h] = a(u,h) - \ell(h) \qquad \forall\, h \in V. \tag{3.10}$$

Proof. Let $u, h \in V$ and $t \in \mathbb{R}$. Using bilinearity of a and linearity of ℓ we compute

$$\mathcal{J}(u+th) - \mathcal{J}(u) = \frac{1}{2}\left(a(u+th, u+th) - a(u,u)\right) - t\ell(h) = t\left(a(u,h) - \ell(h)\right) + \frac{t^2}{2}\,a(h,h).$$

Hence

$$\frac{\mathcal{J}(u+th) - \mathcal{J}(u) - t(a(u,h) - \ell(h))}{\|th\|_{1,\Omega}} = \frac{|t|}{2}\,\frac{|a(h,h)|}{\|h\|_{1,\Omega}} \leq \frac{|t|}{2}\,C_{\mathcal{A}}\,\|h\|_{1,\Omega} \xrightarrow{t\to 0} 0,$$

by boundedness of a as proven in Proposition 3.1. ∎

Proposition 3.4 (First-order optimality condition on convex sets)**.** *A function $u \in K$ minimizes $\mathcal{J}$ over K if and only if it satisfies the variational inequality*

$$\mathcal{J}'(u)[v-u] \geq 0 \qquad \forall\, v \in K. \tag{3.11}$$

Proof. Assume first that $u \in K$ minimizes $\mathcal{J}$ over K and let $v \in K$. By convexity of K we have $u + t(v-u) \in K$ for all $t \in [0,1]$. Define $\phi(t) := \mathcal{J}(u + t(v-u))$. Then ϕ attains its minimum at $t = 0$, hence $\phi'(0^+) \geq 0$. Using the chain rule and Fréchet differentiability of $\mathcal{J}$ we obtain

$$\phi'(0^+) = \mathcal{J}'(u)[v-u] \geq 0.$$

Conversely, assume (3.11). Let $v \in K$ be arbitrary and set $h := v - u$. By Taylor expansion of the quadratic functional $\mathcal{J}$ we have

$$\mathcal{J}(u+h) - \mathcal{J}(u) = \mathcal{J}'(u)[h] + \frac{1}{2}\, a(h,h).$$

By (3.11), $\mathcal{J}'(u)[h] \geq 0$, and since $a(h,h) \geq 0$ we conclude $\mathcal{J}(v) \geq \mathcal{J}(u)$ for all $v \in K$. Hence u is a minimizer over K. ■

Combining Lemma 3.5 and Proposition 3.4, $u \in K$ solves Problem 3.1 if and only if

$$\mathcal{J}'(u)[v-u] = a(u, v-u) - \ell(v-u) \geq 0 \qquad \forall\, v \in K.$$

This is the first weak formulation of Signorini's problem as a variational inequality.

Problem 3.2 (Signorini's Problem – Weak Formulation)**.** *Find $u \in K$ such that*

$$a(u, v-u) \geq \ell(v-u) \qquad \forall\, v \in K.$$

Well-posedness of Problem 3.2 is already ensured, since it is equivalent to the minimization Problem 3.1 in the sense that each solution to the one of the problems is a solution to the other problem (Proposition 3.4). The solution is unique due to Corollary 3.1 with the stability estimate (3.9). Another useful equivalent formulation is the following.

Problem 3.3 (Signorini's Problem – Weak Formulation)**.** *Find $u \in K$ such that*

$$\begin{cases} a(u,u) = \ell(u), \\ a(u,v) \geq \ell(v) \qquad \forall\, v \in K. \end{cases}$$

Proposition 3.5 (Equivalence of Problem 3.2 and Problem 3.3)**.** *Any solution $u \in K$ to Problem 3.2 is also a solution to Problem 3.3 and vice versa.*

Proof. Let $u \in K$ solve Problem 3.3. Then

$$a(u, v-u) = a(u,v) - a(u,u) = a(u,v) - \ell(u) \geq \ell(v) - \ell(u) = \ell(v-u).$$

Assuming that $u \in K$ solves Problem 3.2 and testing $v = 0$

$$a(u,-u) \geq \ell(-u) \qquad \Longleftrightarrow \qquad -a(u,u) \geq -\ell(u) \qquad \Longleftrightarrow \qquad a(u,u) \leq \ell(u).$$

However, testing with $v = 2u$ yields $a(u,u) \geq \ell(u)$ and thus $a(u,u) = \ell(u)$. Furthermore, Problem 3.2 reads

$$a(u,v-u) = a(u,v) - a(u,u) \geq \ell(v-u) = \ell(v) - \ell(u) \qquad \Longrightarrow \qquad a(u,v) \geq a(u,u) + \ell(v) - \ell(u).$$

Plugging in $a(u,u) = \ell(u)$ yields $a(u,v) \geq \ell(v)$. ■

We close this section by providing another stability result on the dependency of solutions to varying source terms as input data.

Proposition 3.6 (Lipschitz dependency)**.** *Let $f_1, f_2 \in L^2(\Omega;\mathbb{R}^d)$ and $F_1, F_2 \in L^2(\Gamma_N;\mathbb{R}^d)$ and let $\ell_i \in V'$ be the corresponding load functionals*

$$\ell_i(v) = \int_\Omega f_i \cdot v \,\mathrm{d}x + \int_{\Gamma_N} F_i \cdot \gamma v \,\mathrm{d}s, \qquad v \in V,\ i = 1,2.$$

Let $u_i \in K$ be the corresponding (unique) minimizers of

$$\mathcal{J}_i(v) := \frac{1}{2}\, a(v,v) - \ell_i(v)$$

over K. Then the solutions depend Lipschitz continuously on the data, i.e.

$$\|u_1 - u_2\|_{1,\Omega} \le \frac{C_K^2}{\alpha_{\mathcal{A}}}\, \|\ell_1 - \ell_2\|_{-1,\Omega} \le \frac{C_K^2\, \max\{1, C_{\mathrm{tr}}\}}{\alpha_{\mathcal{A}}} \left(\|f_1 - f_2\|_{0,\Omega} + \|F_1 - F_2\|_{0,\Gamma_N}\right).$$

Proof. Since $u_i \in K$ minimizes $\mathcal{J}_i$ over the closed convex set K by Proposition 3.4, it satisfies the first-order optimality condition

$$a(u_i, v - u_i) \ge \ell_i(v - u_i) \qquad \forall\, v \in K,\ i = 1, 2. \tag{3.12}$$

Choose $v = u_2$ in (3.12) for $i = 1$ and $v = u_1$ in (3.12) for $i = 2$. This yields

$$\begin{aligned} a(u_1, u_2 - u_1) &\ge \ell_1(u_2 - u_1), \\ a(u_2, u_1 - u_2) &\ge \ell_2(u_1 - u_2). \end{aligned}$$

Adding both inequalities and using bilinearity of a gives

$$a(u_1 - u_2, u_1 - u_2) \le (\ell_1 - \ell_2)(u_1 - u_2).$$

By the V-ellipticity estimate from Proposition 3.1 ii) we have

$$a(w, w) \ge \frac{\alpha_{\mathcal{A}}}{C_K^2}\, \|w\|_{1,\Omega}^2 \qquad \forall\, w \in V.$$

Applying this with $w = u_1 - u_2$ and estimating the right-hand side by the dual norm $\|\cdot\|_{-1,\Omega}$ yields

$$\frac{\alpha_{\mathcal{A}}}{C_K^2}\, \|u_1 - u_2\|_{1,\Omega}^2 \le a(u_1 - u_2, u_1 - u_2) \le \|\ell_1 - \ell_2\|_{-1,\Omega}\, \|u_1 - u_2\|_{1,\Omega}.$$

If $u_1 = u_2$ there is nothing to show. Otherwise we divide by $\|u_1 - u_2\|_{1,\Omega}$ and obtain

$$\|u_1 - u_2\|_{1,\Omega} \le \frac{C_K^2}{\alpha_{\mathcal{A}}}\, \|\ell_1 - \ell_2\|_{-1,\Omega}. \tag{3.13}$$

It remains to bound $\|\ell_1 - \ell_2\|_{-1,\Omega}$ in terms of $f_1 - f_2$ and $F_1 - F_2$. Let $v \in V$ be arbitrary. By Cauchy–Schwarz,

$$|(\ell_1 - \ell_2)(v)| \le \|f_1 - f_2\|_{0,\Omega}\, \|v\|_{0,\Omega} + \|F_1 - F_2\|_{0,\Gamma_N}\, \|\gamma v\|_{0,\Gamma_N}.$$

Using $\|v\|_{0,\Omega} \le \|v\|_{1,\Omega}$ and the trace estimate $\|\gamma v\|_{0,\Gamma_N} \le C_{\mathrm{tr}}\, \|v\|_{1,\Omega}$ we obtain

$$|(\ell_1 - \ell_2)(v)| \le \max\{1, C_{\mathrm{tr}}\} \left(\|f_1 - f_2\|_{0,\Omega} + \|F_1 - F_2\|_{0,\Gamma_N}\right) \|v\|_{1,\Omega}.$$

Dividing by $\|v\|_{1,\Omega}$ and taking the supremum over $0 \ne v \in V$ yields

$$\|\ell_1 - \ell_2\|_{-1,\Omega} \le \max\{1, C_{\mathrm{tr}}\} \left(\|f_1 - f_2\|_{0,\Omega} + \|F_1 - F_2\|_{0,\Gamma_N}\right)$$

Combining this with (3.13) gives the claim. ∎

3.3 Strong Formulation

From the previous weak formulations of Signorini's problem, we know there exists a uniquely determined $u \in K$ that solves the problem. To derive a *strong* PDE formulation, we have to discuss what was $\sigma(u)$n means and in what sense the contact conditions on Γ_C can be described as equalities and inequalities.

3.3.1 Stress Regularity and Boundary Tractions

In the weak setting, the solution only satisfies $u \in K \subset H^1(\Omega;\mathbb{R}^d)$., i.e. $\varepsilon(u) \in L^2(\Omega;\mathrm{Sym}^2(\mathbb{R}^d))$ and, by $\sigma(u) = A : \varepsilon(u)$, we obtain $\sigma(u) \in L^2(\Omega;\mathrm{Sym}^2(\mathbb{R}^d))$. This regularity is sufficient to interpret the internal virtual work

$$a(u,v) = \int_\Omega \sigma(u) : \varepsilon(v)\,\mathrm{d}x,$$

but it is not enough to write the equilibrium equation

$$-\operatorname{div}\sigma(u) = f$$

as an identity in $L^2(\Omega;\mathbb{R}^d)$, since the divergence of an L^2–field is only defined in the sense of distributions. To formulate a strong equilibrium equation in $L^2(\Omega;\mathbb{R}^d)$ and to give a meaning to *boundary tractions*, we assume additional regularity of the stress, $\sigma(u) \in H(\mathrm{div};\Omega)$. This means that $\operatorname{div}\sigma(u) \in L^2(\Omega;\mathbb{R}^d)$, so the equilibrium equation is well–defined. The same regularity is also exactly what is needed to define the boundary traction (or surface force density) associated with τ. Formally, for smooth τ one would write $\tau\,\mathrm{n}$ on Γ where $\mathrm{n} : \Gamma \to \mathbb{R}^d$ is the outward unit normal and $\tau\,\mathrm{n}$ denotes the pointwise matrix-vector product as earlier in (3.2). However, for $\tau \in L^2(\Omega;\mathbb{R}^{d\times d})$ a pointwise boundary trace is not available, so $\tau\,\mathrm{n}$ cannot be defined as an L^2-function on Γ in general. One therefore starts from the classical integration by parts formula.

Proposition 3.7 (Integration by Parts). *If $\tau \in C^1(\overline{\Omega};\mathbb{R}^{d\times d})$ and $v \in C^1(\overline{\Omega};\mathbb{R}^d)$, then*

$$\int_\Omega \tau : \nabla v\,\mathrm{d}x + \int_\Omega (\operatorname{div}\tau)\cdot v\,\mathrm{d}x = \int_\Gamma (\tau\,\mathrm{n})\cdot v\,\mathrm{d}s. \tag{3.14}$$

The key point now is that the left-hand side still makes sense for $\tau \in H(\mathrm{div};\Omega)$ and $v \in H^1(\Omega;\mathbb{R}^d)$ by understanding the differential operators in the weak sense. Hence, for some fixed $\tau \in H(\mathrm{div};\Omega)$ the map

$$F_\tau(v) := \int_\Omega \tau : \nabla v\,\mathrm{d}x + \int_\Omega (\operatorname{div}\tau)\cdot v\,\mathrm{d}x, \qquad v \in H^1(\Omega;\mathbb{R}^d), \tag{3.15}$$

defines a continuous linear functional on $H^1(\Omega;\mathbb{R}^d)$. The crucial observation is that F_τ depends only on the boundary trace γv.

Proposition 3.8 (Dependence of F_τ on the Trace). *The functional F_τ define in (3.15) vanished on $H^1_0(\Omega;\mathbb{R}^d)$, i.e.*

$$F_\tau(w) = 0 \qquad \forall\, w \in H^1_0(\Omega;\mathbb{R}^d).$$

Consequently, F_τ depends only on the trace γv in the sense that

$$\gamma v_1 = \gamma v_2 \quad\Longrightarrow\quad F_\tau(v_1) = F_\tau(v_2) \qquad \forall\, v_1, v_2 \in H^1(\Omega;\mathbb{R}^d).$$

Proof. Let $w \in H^1_0(\Omega;\mathbb{R}^d)$. By density, there exists a sequence $(w_m)_{m\in\mathbb{N}} \subset C_c^\infty(\Omega;\mathbb{R}^d)$ such that

$$w_m \to w \qquad \text{in } H^1(\Omega;\mathbb{R}^d).$$

Since $\tau \in H(\mathrm{div};\Omega)$, we have for every $m \in \mathbb{N}$ that

$$\int_\Omega (\operatorname{div}\tau)\cdot w_m\,\mathrm{d}x = -\int_\Omega \tau : \nabla w_m\,\mathrm{d}x,$$

and therefore

$$F_\tau(w_m) = 0 \qquad \forall\, m \in \mathbb{N}.$$

Passing to the limit $m \to \infty$ and using the continuity of F_τ on $H^1(\Omega;\mathbb{R}^d)$ yields

$$F_\tau(w) = \lim_{m\to\infty} F_\tau(w_m) = 0.$$

Hence F_τ vanishes on $H_0^1(\Omega;\mathbb{R}^d)$. Now let $v_1, v_2 \in H^1(\Omega;\mathbb{R}^d)$ satisfy $\gamma v_1 = \gamma v_2$. Then

$$\gamma(v_1 - v_2) = 0.$$

By the characterization $\ker(\gamma) = H_0^1(\Omega;\mathbb{R}^d)$, we obtain $v_1 - v_2 \in H_0^1(\Omega;\mathbb{R}^d)$. Therefore,

$$F_\tau(v_1) - F_\tau(v_2) = F_\tau(v_1 - v_2) = 0,$$

which proves the claim. ■

Since $\gamma : H^1(\Omega;\mathbb{R}^d) \to H^{1/2}(\Gamma;\mathbb{R}^d)$ is continuous and $\ker(\gamma) = H_0^1(\Omega;\mathbb{R}^d)$, Proposition 3.8 implies that F_τ induces a well-defined continuous linear functional on the trace space $H^{1/2}(\Gamma;\mathbb{R}^d)$. Hence there exists a unique element $\gamma_{\mathrm{n}}(\tau) \in H^{-1/2}(\Gamma;\mathbb{R}^d)$ such that

$$\langle \gamma_{\mathrm{n}}(\tau), \gamma v\rangle_\Gamma = \int_\Omega \tau : \nabla v \,\mathrm{d}x + \int_\Omega (\operatorname{div}\tau)\cdot v \,\mathrm{d}x \qquad \forall\, v \in H^1(\Omega;\mathbb{R}^d). \tag{3.16}$$

The operator $\gamma_{\mathrm{n}} : H(\operatorname{div};\Omega) \to H^{-1/2}(\Gamma;\mathbb{R}^d)$ is called the *normal trace operator* (see [1, Lemma 3.1]). In the smooth case, (3.16) reduces to (3.14), so $\gamma_{\mathrm{n}}(\tau)$ coincides with the classical boundary function $\tau\,\mathrm{n}$. For this reason we write suggestively

$$\gamma_{\mathrm{n}}(\tau) = \tau\,\mathrm{n} \in H^{-1/2}(\Gamma;\mathbb{R}^d).$$

Physically, $\tau\,\mathrm{n}$ is the traction vector (or surface force density) exerted across the boundary with outward normal n, and in particular $t(u) := \sigma(u)\mathrm{n}$ is the surface force induced by the stress field $\sigma(u)$. Finally, since τ is symmetric, $\tau : \nabla v = \tau : \varepsilon(v)$ for all $v \in H^1(\Omega;\mathbb{R}^d)$, so (3.16) can be written in the symmetric form

$$\langle \tau\,\mathrm{n}, \gamma v\rangle_\Gamma = \int_\Omega \tau : \varepsilon(v) \,\mathrm{d}x + \int_\Omega (\operatorname{div}\tau)\cdot v \,\mathrm{d}x \qquad \forall\, (v,\tau) \in H^1(\Omega;\mathbb{R}^d) \times H(\operatorname{div};\Omega).$$

Applying this to $\tau = \sigma(u) \in H(\operatorname{div};\Omega)$ provides

$$\langle \sigma(u)\mathrm{n}, \gamma v\rangle_\Gamma = \int_\Omega \sigma(u) : \varepsilon(v) \,\mathrm{d}x + \int_\Omega \operatorname{div}\sigma(u)\cdot v \,\mathrm{d}x \qquad \forall\, v \in H^1(\Omega;\mathbb{R}^d). \tag{3.17}$$

which will be needed in the strong Signorini conditions.

3.3.2 A Hybrid Formulation

Problem 3.4 (Signorini's Problem – Hybrid Formulation)**.** *For given $f \in L^2(\Omega;\mathbb{R}^d)$ and $F \in L^2(\Gamma_N;\mathbb{R}^d)$, find $u \in K$ such that*

$$\begin{cases} -\operatorname{div}\sigma(u) = f & \text{a.e. in } \Omega, \\ \langle \sigma(u)\mathrm{n}, \gamma u\rangle_\Gamma = \displaystyle\int_{\Gamma_N} F\cdot\gamma u \,\mathrm{d}x, & \\ \langle \sigma(u)\mathrm{n}, \gamma v\rangle_\Gamma \geq \displaystyle\int_{\Gamma_N} F\cdot\gamma v \,\mathrm{d}x & \forall\, v \in K. \end{cases}$$

We close this section by showing, that the hybrid formulation above is equivalent to the previous weak formulations in Problem 3.2 in order to prove that it is indeed a valid formulation of the Signorini problem.

Proposition 3.9 (Equivalence of Hybrid and Weak Formulation). *Every solution $u \in K$ to Problem 3.2 is also a solution to Problem 3.4 and vice versa.*

Proof. Let $u \in K$ solve Problem 3.2, i.e. it holds

$$a(u, v-u) \geq \ell(v-u) \qquad \forall\, v \in K. \tag{3.18}$$

We take a test function $\varphi \in \mathcal{D}(\Omega;\mathbb{R}^d)$. Then $\gamma\varphi = 0$ on Γ and $u \pm t\varphi \in K$ for all $t > 0$. Testing $v := u \pm t\varphi$ in (3.18) yields

$$a(u, \pm t\varphi) \geq \ell(\pm t\varphi) \qquad \Longrightarrow \qquad \begin{cases} a(u,\varphi) \geq \ell(\varphi), \\ a(u,\varphi) \leq \ell(\varphi), \end{cases} \qquad \Longrightarrow \qquad a(u,\varphi) = \ell(\varphi). \tag{3.19}$$

for all $\varphi \in \mathcal{D}(\Omega;\mathbb{R}^d)$. Furthermore since $\gamma\varphi = 0$, we have

$$\int_{\Gamma_N} F \cdot \gamma\varphi \,\mathrm{d}s = 0 \qquad \Longrightarrow \qquad \ell(\varphi) = \int_\Omega f \cdot \varphi \,\mathrm{d}x$$

and with the symmetry $\sigma(u)^\top = \sigma(u)$,

$$a(u,\varphi) = \int_\Omega \sigma(u) : \varepsilon(\varphi)\,\mathrm{d}x = \int_\Omega \sigma(u) : \nabla\varphi \,\mathrm{d}x.$$

Hence, for all $\varphi \in \mathcal{D}(\Omega;\mathbb{R}^d)$, we obtain that (3.19) reads

$$\int_\Omega \sigma(u) : \nabla\varphi \,\mathrm{d}x = \int_\Omega f \cdot \varphi \,\mathrm{d}x.$$

Interpreting $\sigma(u)$ and f as regular distributions, and using the distributional divergence of matrix-valued distributions, this is equivalent to

$$\langle -\operatorname{div}\sigma(u), \varphi\rangle = \langle f, \varphi\rangle \qquad \forall\, \varphi \in \mathcal{D}(\Omega;\mathbb{R}^d).$$

Therefore,

$$-\operatorname{div}\sigma(u) = f \qquad \text{in } \mathcal{D}'(\Omega;\mathbb{R}^d).$$

If, in addition, $\sigma(u) \in H(\operatorname{div};\Omega)$, then $\operatorname{div}\sigma(u) \in L^2(\Omega;\mathbb{R}^d)$. By injectivity of the embedding $L^1_{\mathrm{loc}}(\Omega;\mathbb{R}^d) \hookrightarrow \mathcal{D}'(\Omega;\mathbb{R}^d)$, it follows that

$$-\operatorname{div}\sigma(u) = f \qquad \text{a.e. in } \Omega. \tag{3.20}$$

This is the first condition in Problem 3.4. Now let $v \in K$ be arbitrary and set $w := v - u$. Then by (3.17) and (3.20) we have

$$a(u,w) = -\int_\Omega \operatorname{div}\sigma(u) \cdot w \,\mathrm{d}x + \langle \sigma(u)\mathrm{n},\, \gamma w\rangle_\Gamma = \int_\Omega f \cdot w \,\mathrm{d}x + \langle \sigma(u)\mathrm{n},\, \gamma w\rangle_\Gamma. \tag{3.21}$$

On the other hand, by (3.18) we have

$$a(u,w) \geq \ell(w) = \int_\Omega f \cdot w \,\mathrm{d}x + \int_{\Gamma_N} F \cdot \gamma w \,\mathrm{d}s$$

which by subtracting $\langle f,\, w\rangle$ yields

$$\langle \sigma(u)\mathrm{n},\, \gamma w\rangle_\Gamma \geq \int_{\Gamma_N} F \cdot \gamma w \,\mathrm{d}s \qquad \forall\, v \in K.$$

This is the third condition in Problem 3.4. By Proposition 3.5 the condition (3.18) implies $a(u,u) = \ell(u)$ and $a(u,v) \geq \ell(v)$ for all $v \in K$. Again, with (3.17) and (3.20), we have

$$a(u,u) = \int_\Omega f \cdot u \,\mathrm{d}x + \langle \sigma(u)\mathrm{n}, \gamma u\rangle_\Gamma \qquad \text{and} \qquad \ell(u) = \int_\Omega f \cdot u \,\mathrm{d}x + \int_{\Gamma_N} F \cdot \gamma u \,\mathrm{d}s$$

and thus by $a(u,u) = \ell(u)$ and subtracting $\langle f, u\rangle$,

$$\langle \sigma(u)\mathrm{n}, \gamma u\rangle_\Gamma = \int_{\Gamma_N} F \cdot \gamma u \,\mathrm{d}s.$$

This is the second condition in Problem 3.4. Now we assume that $u \in K$ solves Problem 3.4. Let $v \in K$ be arbitrary and $w := v - u$. Subtracting the boundary conditions in Problem 3.4 yields

$$\langle \sigma(u)\mathrm{n},\, \gamma w\rangle_\Gamma = \langle \sigma(u)\mathrm{n},\, \gamma v\rangle_\Gamma - \langle \sigma(u)\mathrm{n},\, \gamma u\rangle_\Gamma \geq \int_{\Gamma_N} F \cdot (\gamma v - \gamma u) \,\mathrm{d}s = \int_{\Gamma_N} F \cdot \gamma w \,\mathrm{d}s.$$

With (3.21) is holds

$$a(u,w) = \int_\Omega f \cdot w \,\mathrm{d}x + \langle \sigma(u)\mathrm{n}, \gamma w\rangle_\Gamma \geq \int_\Omega f \cdot w \,\mathrm{d}x + \int_{\Gamma_N} F \cdot \gamma w \,\mathrm{d}s = \ell(w) = \ell(v-u).$$

This is exactly Problem 3.2. ■

3.3.3 Normal/Tangential Splitting and Contact Trace Spaces

The hybrid formulation derived above still encodes the contact interaction through a single boundary traction functional acting on traces. In order to recover the Signorini contact conditions on Γ_C in a meaningful weak sense, we now split traces and tractions into normal and tangential components and introduce the corresponding trace spaces on the contact boundary. Recall that for $v \in V$, the trace $\gamma v|_{\Gamma_C}$ is well-defined. We define the contact trace space by

$$W_C := \{\gamma v|_{\Gamma_C} \mid v \in V\} \subset H^{1/2}(\Gamma_C; \mathbb{R}^d).$$

For $w \in W_C$, we define its normal and tangential components by

$$w_\mathrm{n} := w \cdot \mathrm{n}, \qquad w_t := w - (w \cdot \mathrm{n})\mathrm{n}.$$

Accordingly, we introduce the normal and tangential trace spaces

$$W_{C,\mathrm{n}} := \{w \cdot \mathrm{n} \mid w \in W_C\}, \qquad W_{C,t} := \{w \in W_C \mid w \cdot \mathrm{n} = 0\}.$$

Hence every $w \in W_C$ admits the decomposition

$$w = w_\mathrm{n}\mathrm{n} + w_t \in W_{C,\mathrm{n}}\,\mathrm{n} \oplus W_{C,t}$$

We endow W_C, $W_{C,\mathrm{n}}$ and $W_{C,t}$ with the quotient norms induced by V, namely

$$\|w\|_{W_C} := \inf\{\|v\|_{1,\Omega} \mid v \in V,\ \gamma v|_{\Gamma_C} = w\}, \qquad w \in W_C, \tag{3.22}$$

and

$$\|\eta\|_{W_{C,\mathrm{n}}} := \inf\{\|v\|_{1,\Omega} \mid v \in V,\ (\gamma v|_{\Gamma_C}) \cdot \mathrm{n} = \eta\}, \qquad \eta \in W_{C,\mathrm{n}}, \tag{3.23}$$

and

$$\|\zeta\|_{W_{C,t}} := \inf\{\|v\|_{1,\Omega} \mid v \in V,\ (\gamma v|_{\Gamma_C})_t = \zeta\}, \qquad \zeta \in W_{C,t}. \tag{3.24}$$

We denote by W'_C, $W'_{C,\mathrm{n}}$ and $W'_{C,t}$ the corresponding dual spaces, equipped with the operator norms. Their duality pairings are denoted by $\langle\cdot,\cdot\rangle_C$. Whenever no confusion can arise, we use the same notation $\langle\cdot,\cdot\rangle_C$ for the pairings on $W'_C \times W_C$, $W'_{C,\mathrm{n}} \times W_{C,\mathrm{n}}$ and $W'_{C,t} \times W_{C,t}$. In the hybrid formulation, the boundary traction on Γ_C is understood as a bounded linear functional on W_C. Hence, for a contact traction $\tau \in W_C'$, one can define normal and tangential components

$$\tau_{\mathrm{n}} \in W_{C,\mathrm{n}}', \qquad \tau_t \in W_{C,t}', \tag{3.25}$$

such that, for every $w \in W_C$ with decomposition $w = w_{\mathrm{n}}\mathrm{n} + w_t$,

$$\langle \tau, w\rangle_C = \langle \tau_{\mathrm{n}}, w_{\mathrm{n}}\rangle_C + \langle \tau_t, w_t\rangle_C. \tag{3.26}$$

To express the Signorini contact condition on the normal component in dual form, we introduce the cone of admissible normal traces

$$W_{C,\mathrm{n}}^- := \{\eta \in W_{C,\mathrm{n}} \mid \eta \le 0 \text{ a.e. on } \Gamma_C\}, \tag{3.27}$$

and its dual cone of weakly nonpositive normal contact forces

$$\Lambda_C := \left\{\lambda \in W_{C,\mathrm{n}}' \;\middle|\; \langle \lambda, \eta\rangle_C \ge 0 \quad \forall\, \eta \in W_{C,\mathrm{n}}^-\right\}. \tag{3.28}$$

With these notations, the boundary inequality in the hybrid formulation can be decomposed into normal and tangential contributions on Γ_C. This allows us to state the frictionless Signorini contact conditions as a KKT-type system in the dual spaces $W_{C,\mathrm{n}}'$ and $W_{C,t}'$.

3.3.4 KKT Contact Conditions

Under the assumption of having no Neumann boundary part, $|\Gamma_N|$, we can make a strong formulation of Signorini's Problem. We denote the boundary stress by

$$\sigma(u)\mathrm{n} = \sigma_{\mathrm{n}}(u)\mathrm{n} + \sigma_t(u)$$

Problem 3.5 (Signorini's Problem – Strong Formulation)**.** *For given $f \in L^2(\Omega;\mathbb{R}^d)$, find $u \in V$ such that*

$$\begin{cases} -\operatorname{div}\sigma(u) = f & \text{a.e. in } \Omega, \\ \sigma_t(u) = 0 & \text{in } W_{C,t}', \\ u_{\mathrm{n}} \le 0 & \text{a.e. on } \Gamma_C, \\ \sigma_{\mathrm{n}}(u) \in \Lambda_C & \\ \langle \sigma_{\mathrm{n}},\, u_{\mathrm{n}}\rangle_{\Gamma_C} = 0, & \end{cases}$$

Proposition 3.10 (Equivalence of Strong and Weak Formulation)**.** *Suppose that $|\Gamma_N| = 0$. Then every solution to Problem 3.5 is also a solution to Problem 3.2 and vice versa.*

Proof. ■

3.3.5 Classical Strong From under Additinal Regularity

Finally, under higher regularity assumptions on u, we can formulate Signorini's problem in a natural strong way. Here $|\Gamma_N| > 0$ is allowed.

Problem 3.6 (Signorini's Problem – Strong Formulation). *For given $f \in L^2(\Omega;\mathbb{R}^d)$ and $F \in L^2(\Gamma_N;\mathbb{R}^d)$ find $u \in H^s(\Omega;\mathbb{R}^d)$ with $s > 3/2$ such that the small strain elasticity equations*

$$\begin{cases} -\operatorname{div}\sigma(u) = f & \textit{in } \Omega, \\ \sigma(u) = \mathcal{A} : \varepsilon(u) & \textit{in } \Omega, \\ u = 0 & \textit{on } \Gamma_D, \\ \sigma(u)\mathrm{n} = F & \textit{on } \Gamma_N, \end{cases}$$

and the contact conditions

$$\begin{cases} u_{\mathrm{n}} \leq 0 & \textit{on } \Gamma_C, \\ \sigma_{\mathrm{n}}(u) \leq 0 & \textit{on } \Gamma_C, \\ \sigma_n(u)\, u_{\mathrm{n}} = 0 & \textit{on } \Gamma_C, \\ \sigma_t(u) = 0 & \textit{on } \Gamma_C, \end{cases}$$

hold.

Proposition 3.11. *Equivalence of Strong and Weak Formulation If $u \in H^s(\Omega;\mathbb{R}^d)$ with $s > 3/2$ is a solution to Problem 3.6, then it is also a solution to Problem 3.2 and vice versa.*

Proof. ■

3.4 Mixed Formulation

In the primal setting, Signorini's problem is posed as a variational inequality on the convex set $K \subset V$. In the mixed approach we introduce an additional unknown λ on the contact boundary, which represents the weak normal contact traction. This leads to a saddle-point structure in the variables (u, λ), where the nonpenetration constraint is encoded by the cone condition $\lambda \in \Lambda_C$.

3.4.1 Mixed Weak Formulation as a Saddlepoint Problem

We keep the energy functional

$$\mathcal{J}(v) = \frac{1}{2}a(v, v) - \ell(v), \qquad v \in V$$

from (3.7) and use the normal trace space $W_{C,\mathrm{n}}$ and the dual cone $\Lambda_C \subset W'_{C,\mathrm{n}}$ introduced in (3.28). For $v \in V$ and $\lambda \in W'_{C,\mathrm{n}}$ we define the coupling form

$$b(v, \lambda) := \langle \lambda,\, (\gamma v|_{\Gamma_C}) \cdot \mathrm{n} \rangle_C\,. \tag{3.29}$$

We then introduce the Lagrangian on $V \times \Lambda_C$ by

$$\mathcal{L}(v, \lambda) := \mathcal{J}(v) - b(v, \lambda). \tag{3.30}$$

Problem 3.7 (Signorini's Problem – Mixed weak formulation). *Find $(u, \lambda) \in V \times \Lambda_C$ such that*

$$\begin{cases} a(u, v) - b(v, \lambda) = \ell(v) & \forall\, v \in V, \\ b(u, \mu - \lambda) \geq 0 & \forall\, \mu \in \Lambda_C. \end{cases}$$

The first condition (**??**) corresponds to bulk equilibrium with an additional boundary reaction term encoded by λ. The second condition (**??**) is a variational inequality on the cone Λ_C and represents the complementarity structure of frictionless contact.

Remark 3.1 (Saddlepoint property)**.** *A pair $(u, \lambda) \in V \times \Lambda_C$ solves Problem 3.7 if and only if it satisfies the saddlepoint inequality*

$$\mathcal{L}(u, \mu) \leq \mathcal{L}(u, \lambda) \leq \mathcal{L}(v, \lambda) \quad \forall\, (v, \mu) \in V \times \Lambda_C. \tag{3.31}$$

3.4.2 Well–posedness and inf–sup condition

We briefly explain the role of the inf–sup condition for the coupling form $b(\cdot, \cdot)$ in (3.29). Define the operators $A : V \to V'$ and $B : V \to W_{C,\mathrm{n}}$ by

$$(Au)(v) := a(u, v), \quad Bv := (\gamma v|_{\Gamma_C}) \cdot \mathrm{n}. \tag{3.32}$$

Then (**??**) can be written as

$$Au - B^*\lambda = \ell \quad \text{in } V', \tag{3.33}$$

where $B^* : W'_{C,\mathrm{n}} \to V'$ is the adjoint operator given by

$$(B^*\lambda)(v) = b(v, \lambda) \quad \forall\, v \in V. \tag{3.34}$$

In the purely linear saddlepoint setting (without the cone constraint $\lambda \in \Lambda_C$), well–posedness of the block system is governed by the inf–sup condition

$$\inf_{\lambda \in W'_{C,\mathrm{n}} \setminus \{0\}} \sup_{v \in V \setminus \{0\}} \frac{b(v, \lambda)}{\|v\|_{1,\Omega}\, \|\lambda\|_{W'_{C,\mathrm{n}}}} \geq \beta > 0. \tag{3.35}$$

Equivalently, (3.35) can be written as

$$\|B^*\lambda\|_{V'} \geq \beta\, \|\lambda\|_{W'_{C,\mathrm{n}}} \quad \forall\, \lambda \in W'_{C,\mathrm{n}}. \tag{3.36}$$

This expresses stability of the coupling between u and λ and, in particular, implies uniqueness of the multiplier once u is fixed.

Proposition 3.12 (Uniqueness of the multiplier)**.** *Assume that* (3.35) *holds. If (u, λ_1) and (u, λ_2) satisfy* (**??**) *with the same $u \in V$, then $\lambda_1 = \lambda_2$ in $W'_{C,\mathrm{n}}$.*

Proof. Subtracting (**??**) for λ_1 and λ_2 yields

$$b(v, \lambda_1 - \lambda_2) = 0 \quad \forall\, v \in V. \tag{3.37}$$

Hence $(B^*(\lambda_1 - \lambda_2))(v) = 0$ for all $v \in V$, i.e. $B^*(\lambda_1 - \lambda_2) = 0$ in V'. Using (3.36) gives $\lambda_1 - \lambda_2 = 0$. ∎

Finally, note that well–posedness of the mixed Signorini formulation also follows from its equivalence to the primal (or hybrid) formulation established earlier. Nevertheless, the inf–sup viewpoint is useful since it explains the saddle-point structure and the stability of the multiplier.

4 Outlook

A References

[1] Franz Chouly, Patrick Hild, and Yves Renard. *Finite Element Approximation of Contact and Friction in Elasticity*, volume 48 of *Advances in Continuum Mechanics*. Springer International Publishing, 2023.

B List of Figures